

Whole-brain TRAP —candidate stories

A) Dual-role craving (Active), B) Passive withdrawal (patient-relevant)

Two findings — why keep both

- FINDING A — ORBm & BMAp: dual-mode craving regions (Active >> Passive).
- Opioid seeking at Post + negative-affect craving persists in Withdrawal. Passive did not show any change
- FINDING B — COAa: Passive > Active at Post and Withdrawal (patient-relevant). Best anatomical fit in cluster 3 for cue/context + withdrawal-state → generalized motivation candidate

Why finding A and finding B targets?

To solve the problem: Opioid seeking vs general seeking (To exclude general seeking possibility)

Why ORBm/BMAp?

They stay **Active** >> **Passive** with Passive flat. Shared general-seeking / context-only accounts predict similar Passive trajectories; we don't see that. So these regions **favor opioid-related Active seeking** and **partly exclude** pure general-seeking circuits.

Why not behavior-matched regions for Finding A?

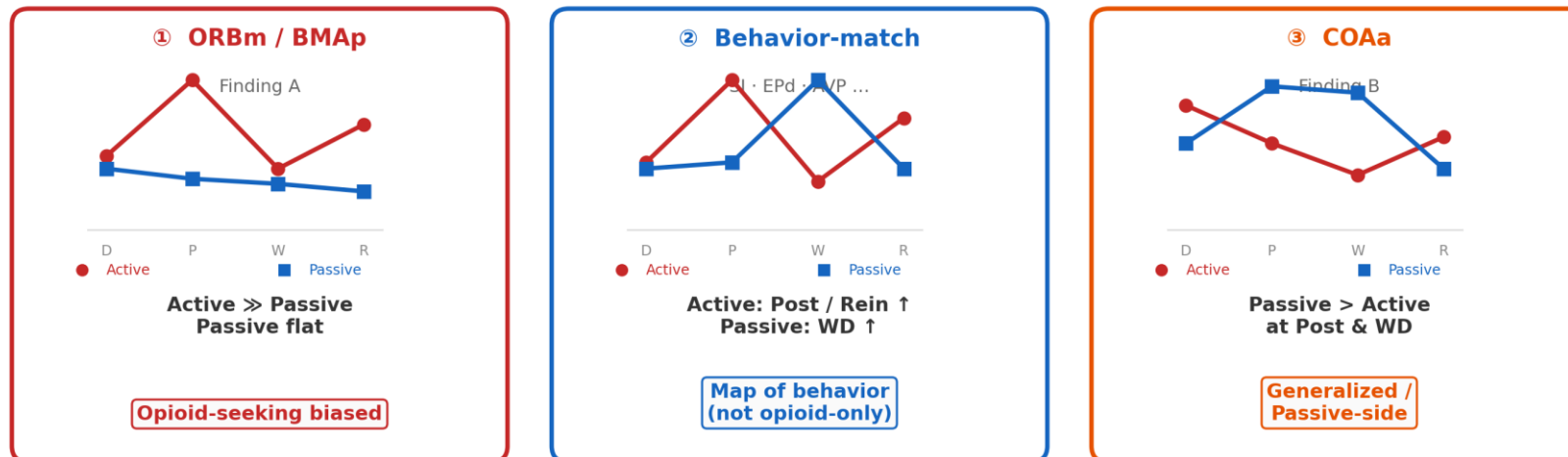
Regions that match behavior patten (Active Post/Rein high; Passive WD high, often > Active) change in **both** delivery modes. That means they track **opioid craving and Passive generalized motivation** — useful as a map, but not opioid-selective.

Why COAa?

Passive > Active at Post and Withdrawal — the epochs of context + withdrawal state in our schematic. COAa is a **generalized / Passive-side** candidate: manipulating it may affect **broad seeking** (e.g. sucrose), whereas ORBm/BMAp should spare that more. WD is not pure sucrose, so this remains a **testable prediction**, not a proven dissociation.

Three circuit types

Active = opioid + seeking · Passive = same drug, no seeking



Analysis pipeline Roadmap — why cluster first, then zoom in

Funnel logic: hundreds of regions → 4 clusters → 2 behaviour-relevant clusters → 7 regions → zoom-in-> only 2 brain regions : ORBm, BMAp

Problem: the brain has hundreds of regions. Testing each one individually is under-powered and hard to interpret (multiple-comparisons, no structure).

Step 1 — FILTER by clustering: pool all forebrain regions (L/R-averaged) and group them by their Active-vs-Passive activity fingerprint across all 4 phases (universal PCA + k-means, K=4). Hundreds of regions → 4 functional clusters.

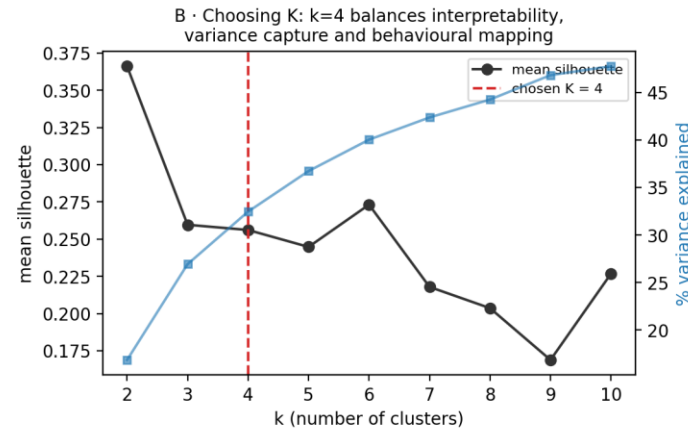
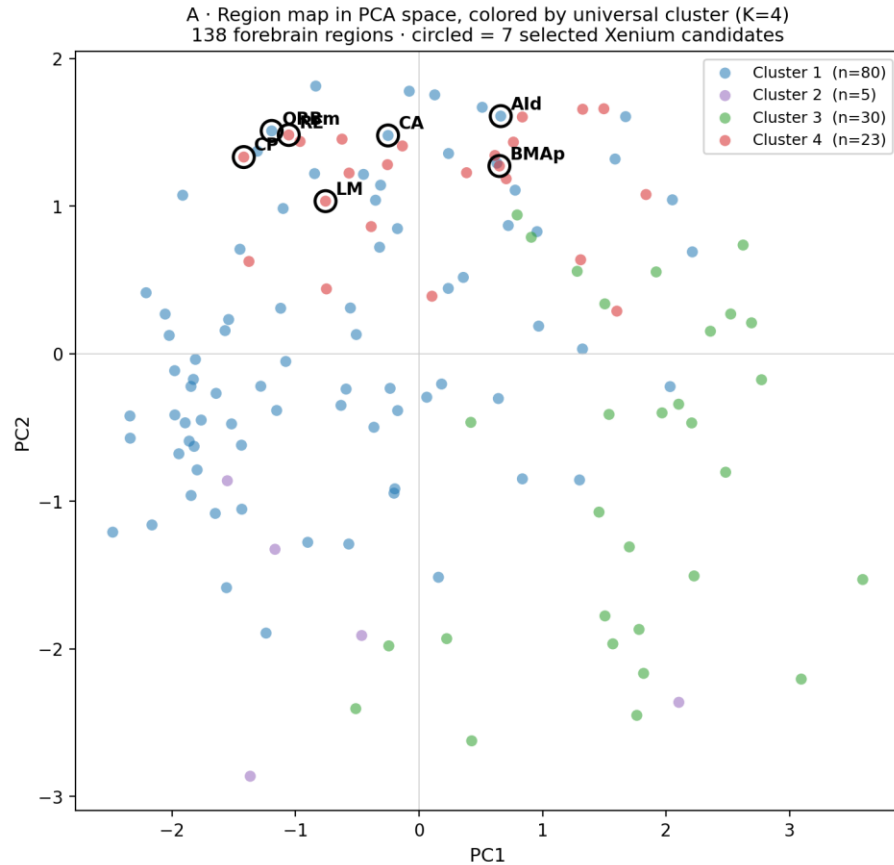
Step 2 — KEEP behaviourally-aligned clusters: only clusters 1 & 4 reproduce the motivation timeline (Active > Passive at Post & Reinstatement). → shortlist of candidate regions.

Step 3 — SELECT **active always higher than passive regions**: BMAp, LM, RE, CP (cluster 4) + ORBm, CA, Ald (cluster 1) = 7 regions, agreeing with an independent top-N ranking.

Step 4 — ZOOM IN: for each of the 7 regions, look at density AND cell count across phases, Active vs Passive, showing every individual mouse. **After ZOOM IN only ORBm and BMAp survived** while others are less promising

Fig 1 · How clustering was done + what clusters look like

Fig 1 · Universal PCA + k-means clustering of whole-brain TRAP activity



C · How the clustering was done

- Input: whole-brain TRAP density (cells/mm³), L/R-averaged, forebrain (no brainstem/cerebellum/fiber).
- Each region z-scored across all samples, pooled over the 4 phases (During, Post, Withdrawal, Reinstatement) → a single 'universal' activity fingerprint per region.
- k-means (K=4, squared-Euclidean, 50 restarts, seed 42) groups regions with similar cross-phase A-vs-P profiles.
- PCA (same matrix) used only for 2-D visualization (A).
- Universal labels are fixed across phases, so a region's cluster is comparable in every phase.

Method (Fig 1)

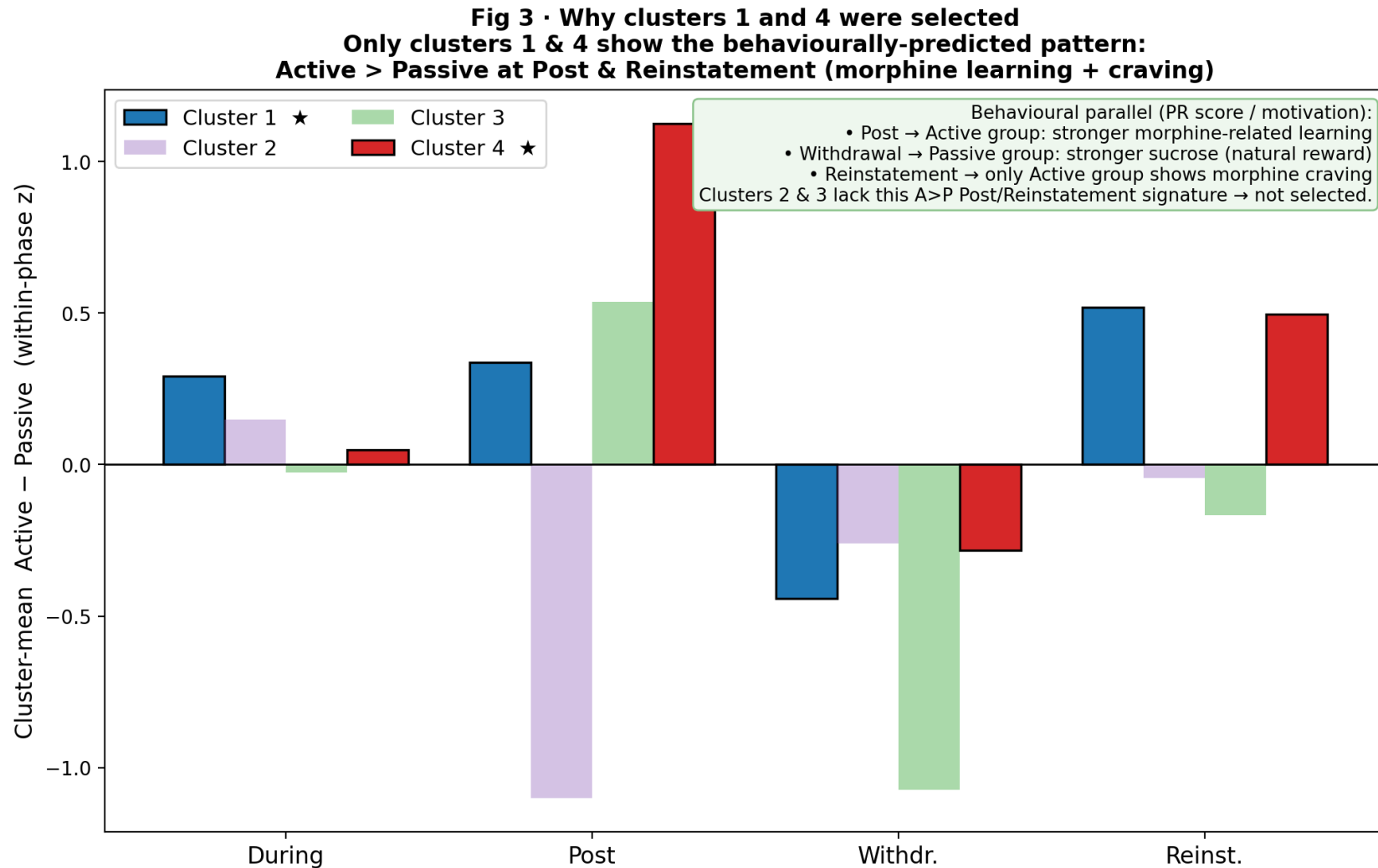
A: 138 forebrain regions in PCA space, colored by cluster (K=4); circled = 7 selected.

B: k was chosen at 4 (balances interpretability, variance, behaviour mapping).

C: each region z-scored across samples (4 phases pooled) → k-means (sq-Euclidean, 50 restarts).

PCA is only for 2-D display; clusters use the full profile.

Fig 3 · Why clusters 1 and 4 were selected



Selection of clusters (Fig 3)

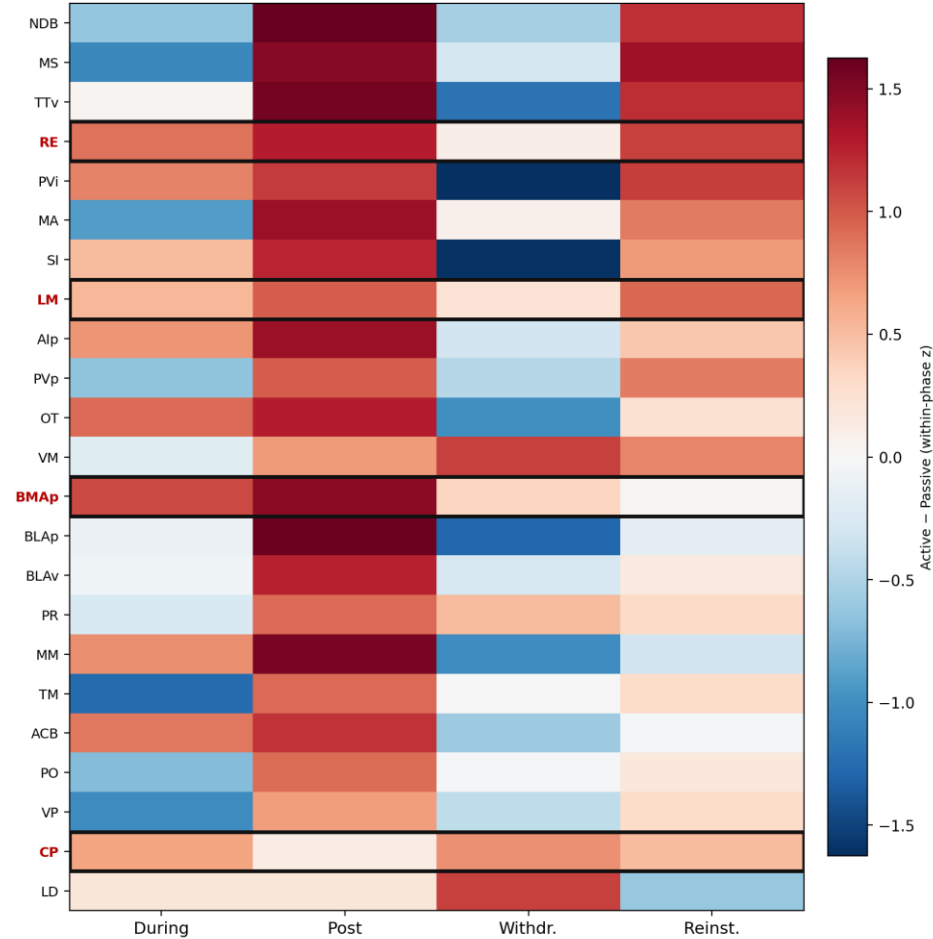
Per-phase bars of cluster-mean Active – Passive.

Only clusters 1 & 4 show the predicted pattern (A>P at Post & Reinstatement).

Post = Active morphine learning; Withdrawal = Passive sucrose; Reinstatement = Active craving.

Fig 4 · Cluster 4 — why BMAp, LM, RE, CP

Fig 4 · Cluster 4 (n=23, amygdala/striatal family): region selection
rows ranked by Post+Reinstatement Active–Passive · boxed = selected (BMAp, LM, RE, CP)



Within cluster 4 (Fig 4)

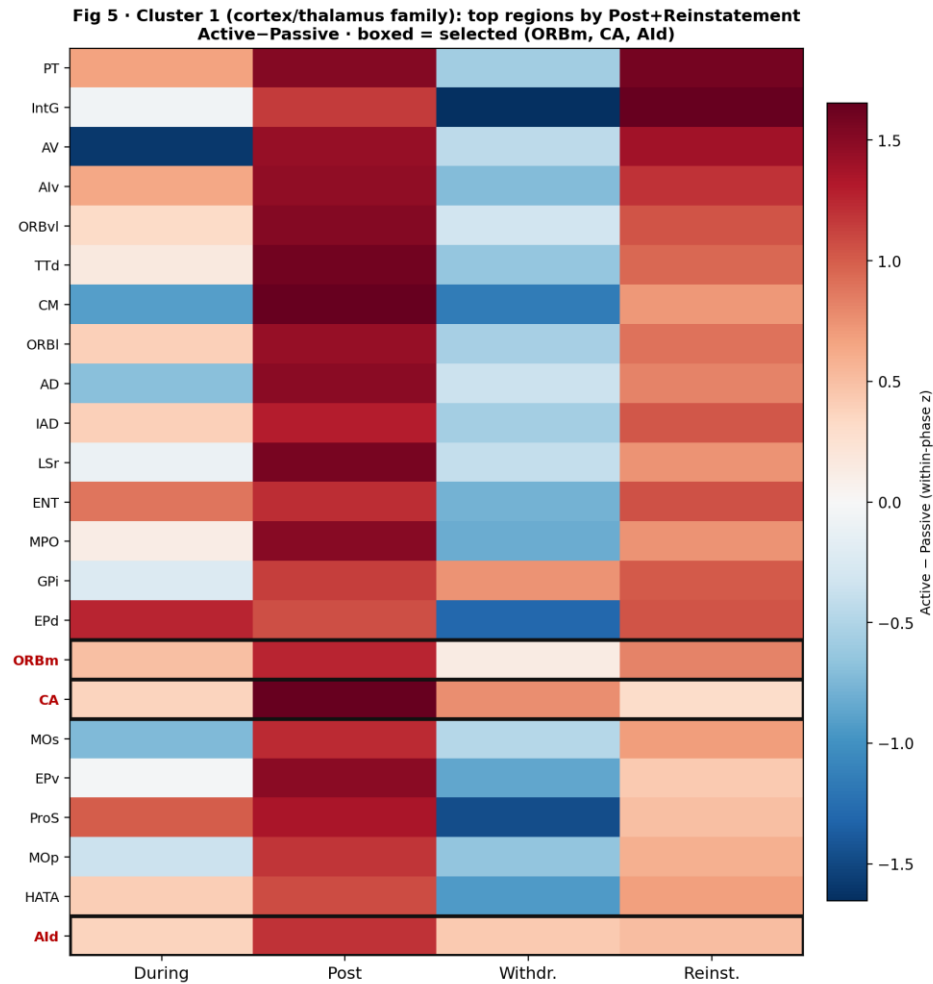
cluster-4 regions × phase heatmap of Active – Passive; boxed = selected.

Selection = A>P in craving phases

- FINDING A -mixed craving regions (Active >> Passive).

Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 4; (2) Active > Passive in the craving-relevant phases (Post/Reinstatement); (3) canonical addiction-circuit node with a defined manipulation target. Other statistically-elevated cluster-4 regions (e.g. NDB, MS, TTv) were deprioritised as less specific / harder to target. BMAp, LM, RE, CP satisfy all three and agree with the independent top-N ranking.

Fig 5 · Cluster 1 — why ORBm, CA, Ald



Within cluster 1 (Fig 5)

Top cluster-1 regions by Post+Reinstatement Active – Passive; boxed = selected.

- FINDING A -mixed craving regions (Active >> Passive).

Selection rule (3 criteria, not score alone): (1) member of behaviourally-aligned cluster 1; (2) Active > Passive at Post/Reinstatement; (3) canonical addiction-relevant cortical/limbic node. Higher-scoring but less interpretable nodes (e.g. PT, IntG, AV thalamic nuclei) were deprioritised. ORBm, CA, Ald satisfy all three and agree with the independent top-N ranking.

Part 2 · Zoom-in on each selected region (group level)

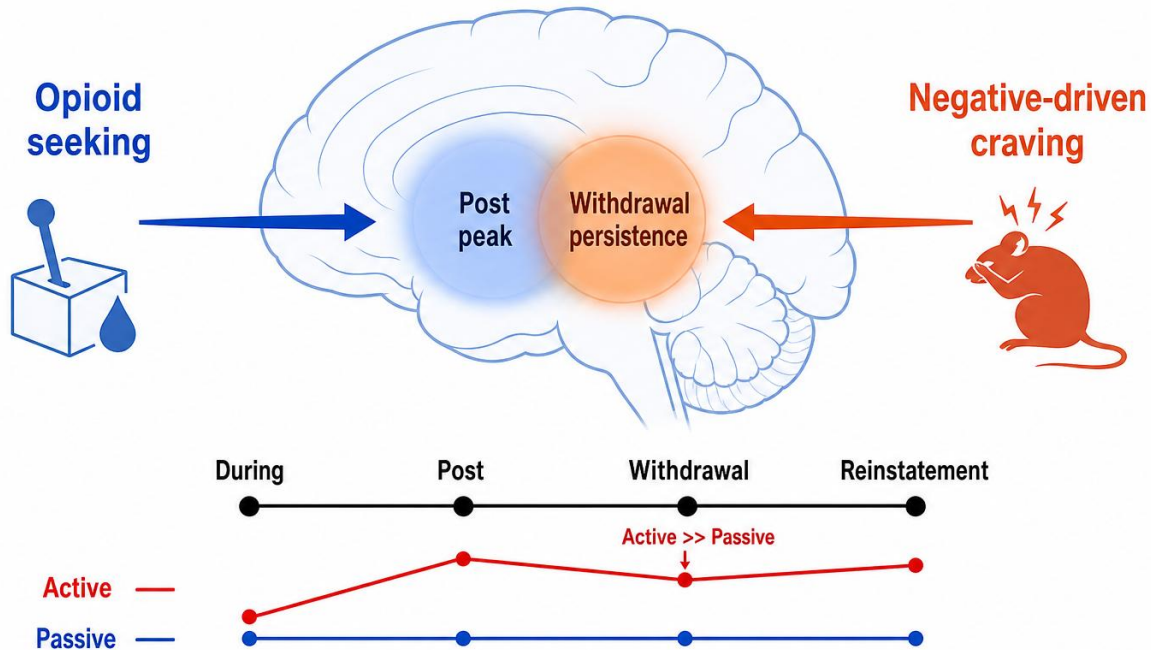
density + cell count · Active vs Passive · individual mice shown

For each of the 7 regions I plot BOTH density (cells/mm³) and cell count across the 4 phases, separately for Active and Passive.

Finding A — Why Active >> Passive even in Withdrawal?

Target = mixed regions: opioid seeking AND negative-affect craving
(Active >> Passive through Withdrawal) → ORBm, BMAp

Finding A — Dual-mode craving regions



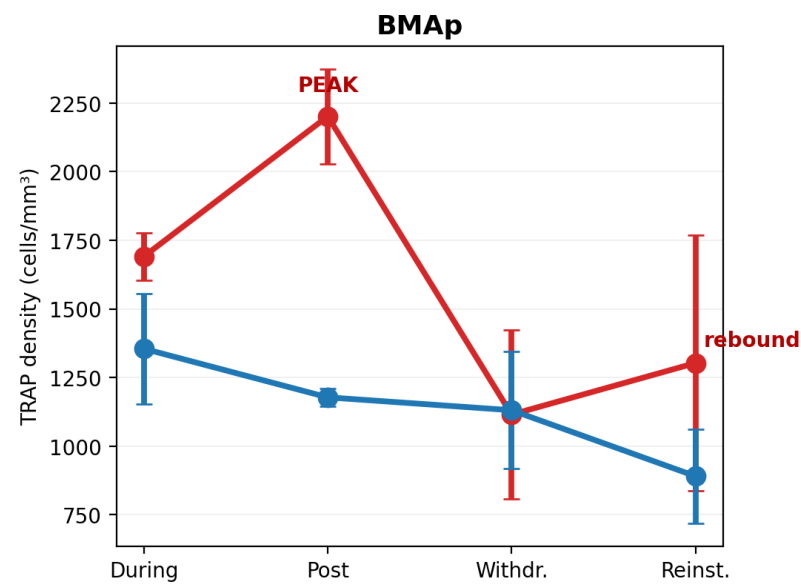
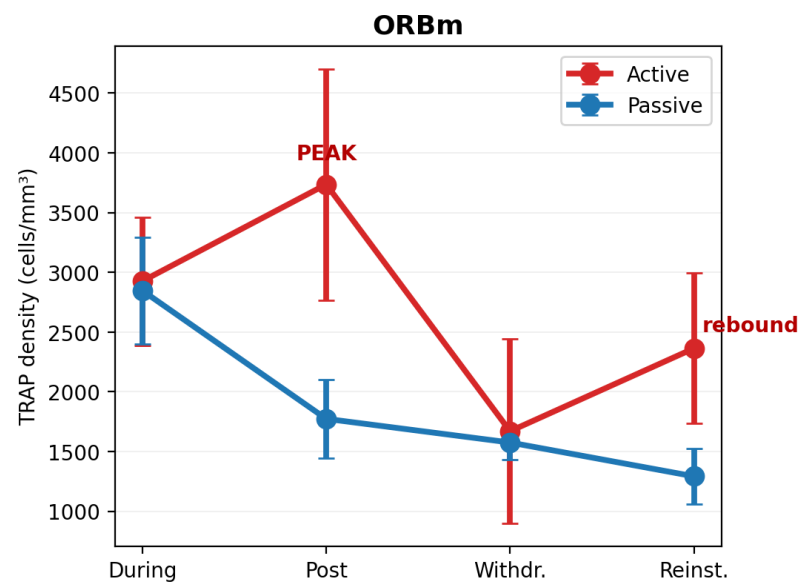
ORBm & BMAp = dual-mode: opioid seeking + negative-affect craving (Active only)

- Not just Post craving — Withdrawal matters for incubation.
- Negative affect during Withdrawal can drive craving increase.
- Target = mixed regions: opioid seeking + negative-driven craving.
- Active >> Passive through Withdrawal (Passive flat).
- → ORBm, BMAp = dual-role craving regions.

Finding A — ORBm & BMAp (raw density)

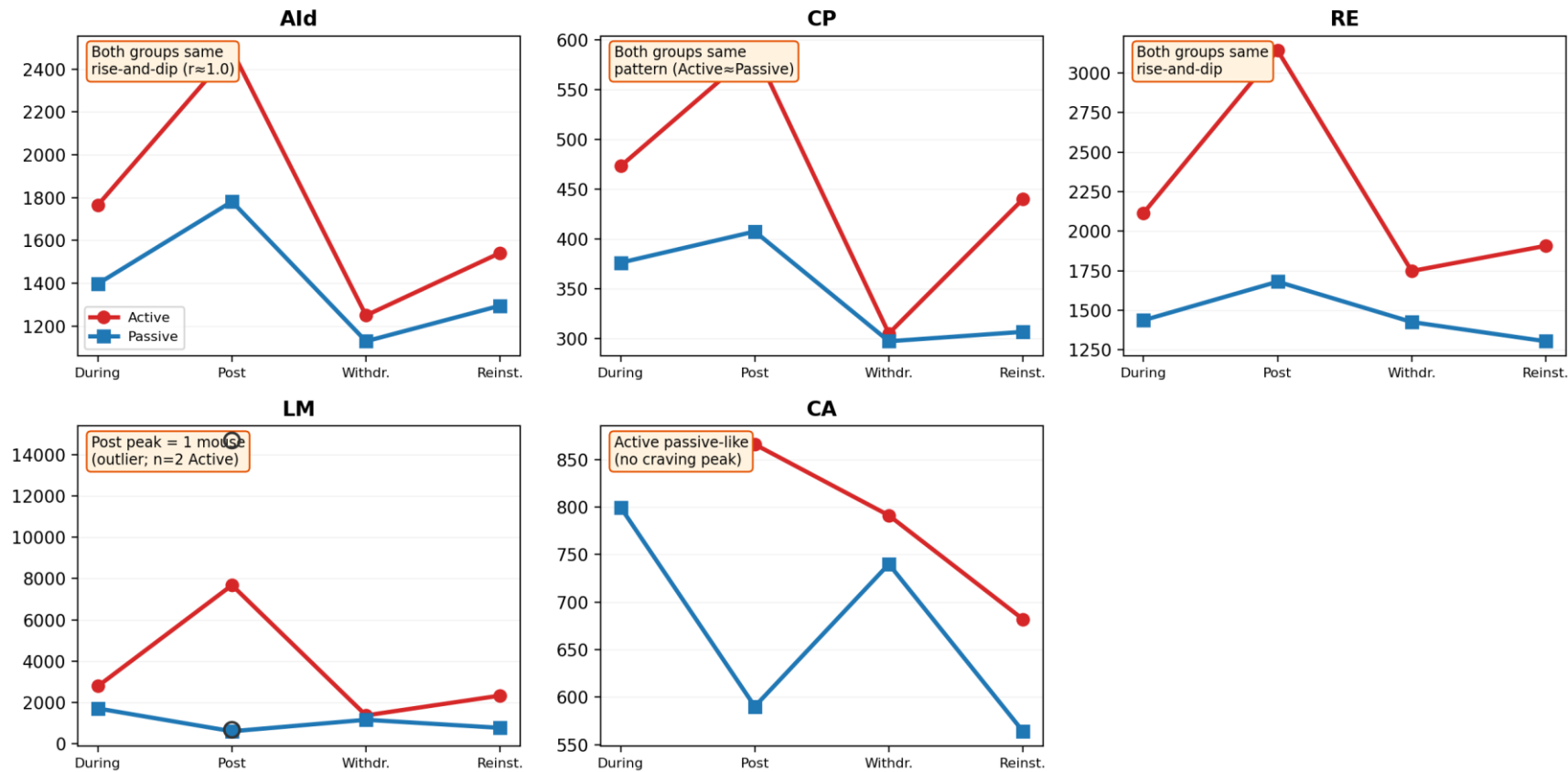
Active-specific: Active > Passive, peaks at Post, rebounds at Reinstatement — Passive stays flat

- Active PEAKS at Post.
- Dips at Withdrawal but stays >> Passive.
- REBOUNDS at Reinstatement.
- Passive flat — no craving pattern.



Why not the other original-7 regions?

Deprioritized from original 7 — why they do NOT isolate Active craving

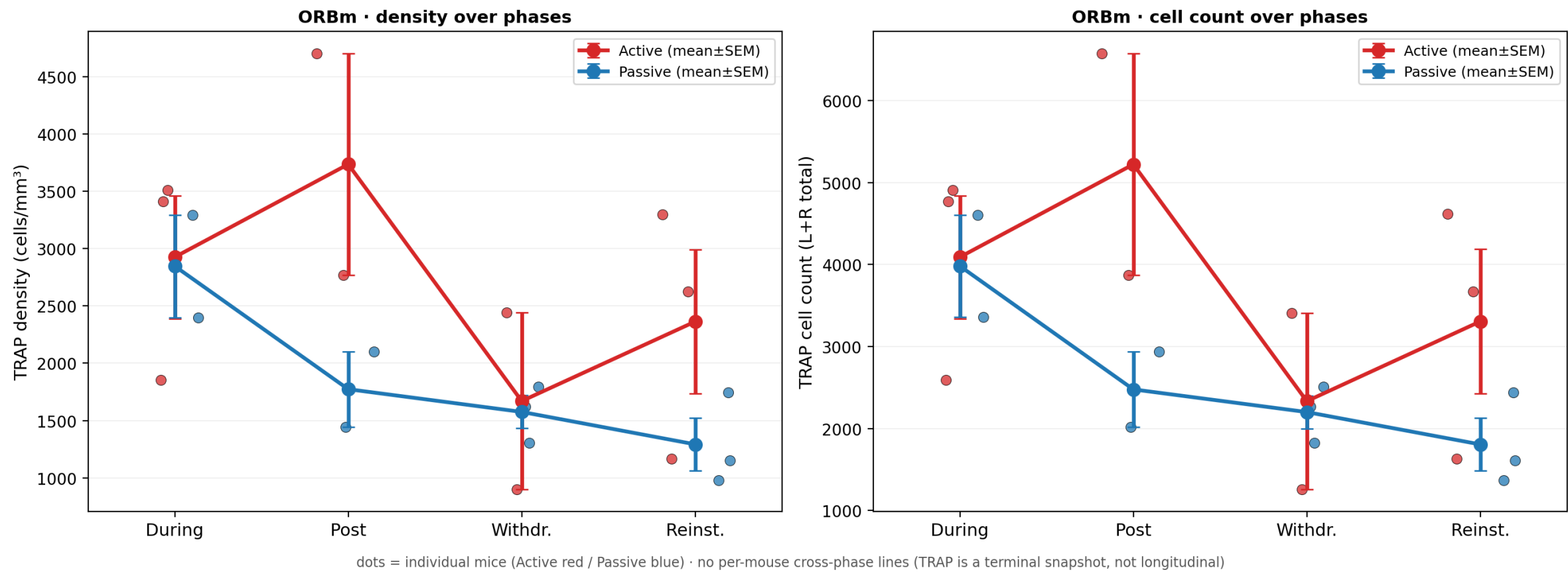


Raw group-mean density (cells/mm³) · terminal design · LM circles = individual Active Post mice ($n=2$)

- Ald, CP, RE: Active $\approx$ Passive — same rise-and-dip (cannot isolate craving).
- Ald = weakest: trajectories nearly identical ($r \approx 1.0$).
- CA: Active is passive-like — no clean Post craving peak.
- LM: Post “peak” driven by 1 mouse ($n=2$ Active at Post).
- → ORBm & BMAp kept as clean Active-only dual-mode regions.

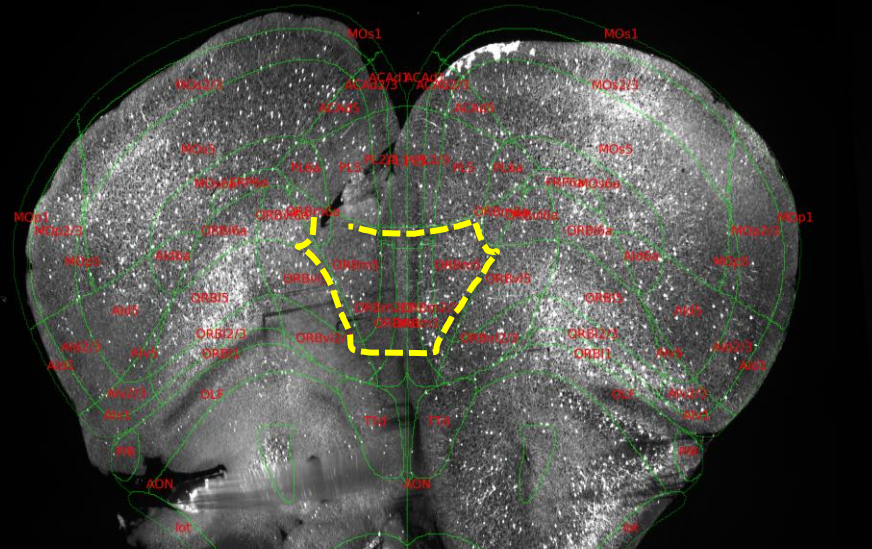
Finding A — ORBm zoom-in (density & cell count)

ORBm (Cluster 1) — individual mice (dots) + group mean±SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice

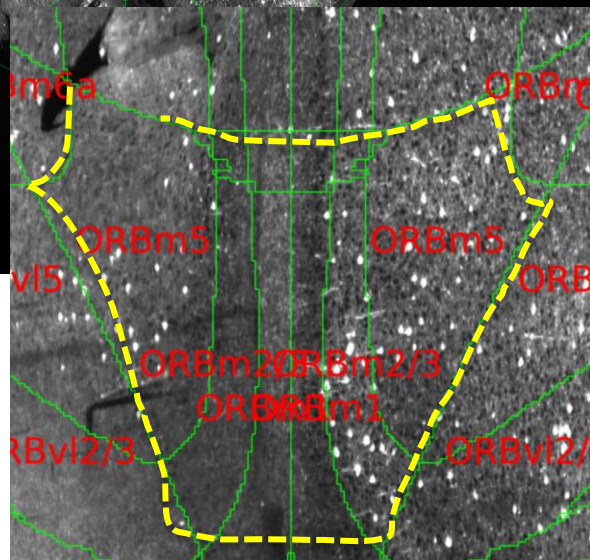


Group trajectories. Active peaks Post, dips Withdrawal, rebounds; Passive flat.

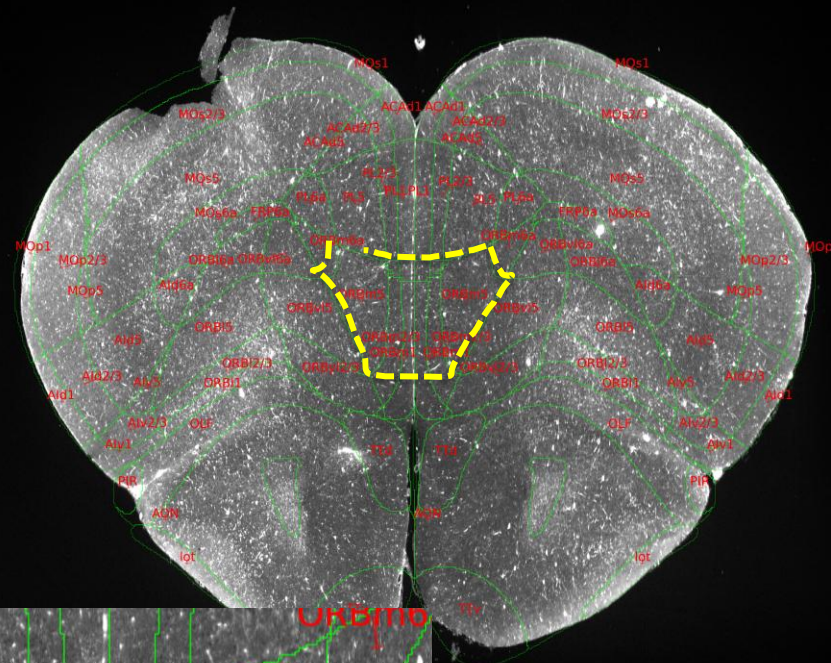
Post



active



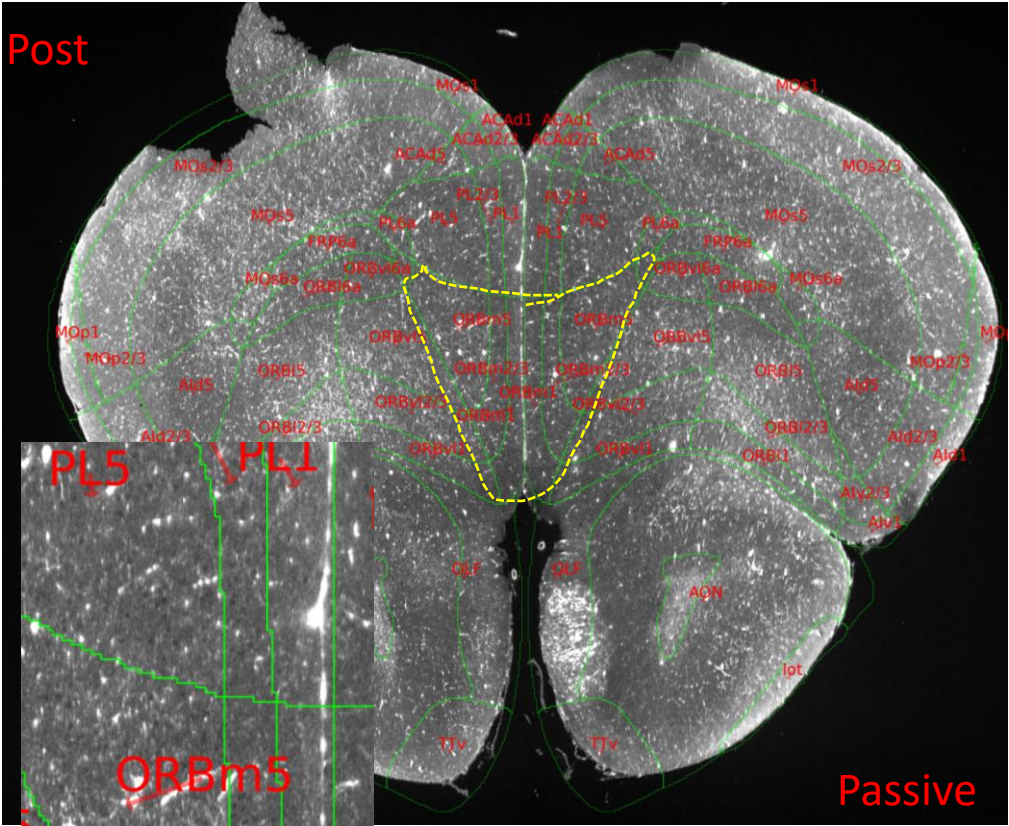
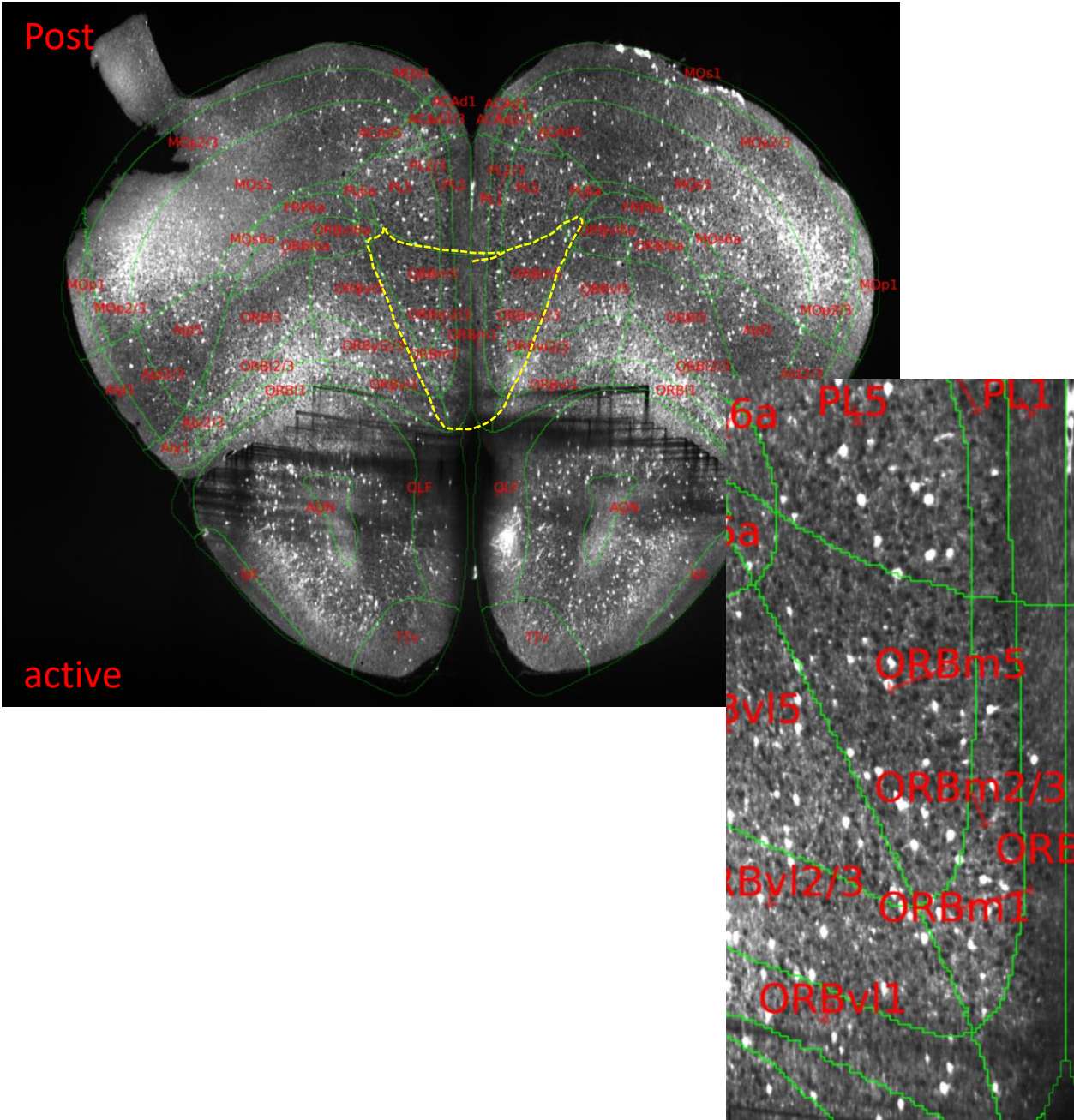
Post

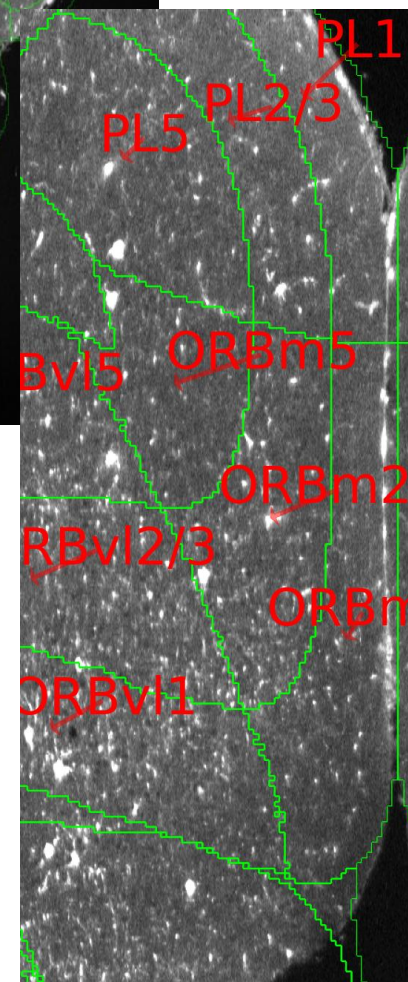
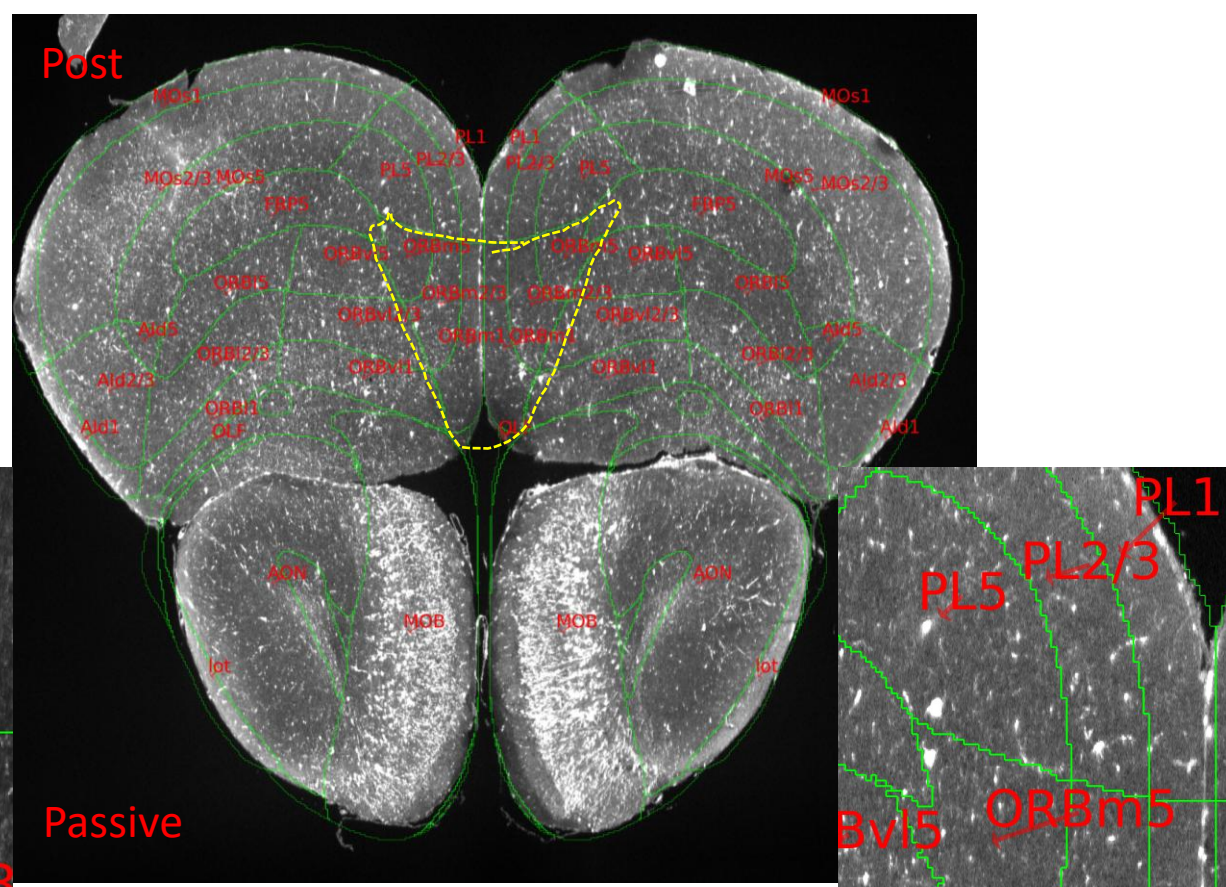
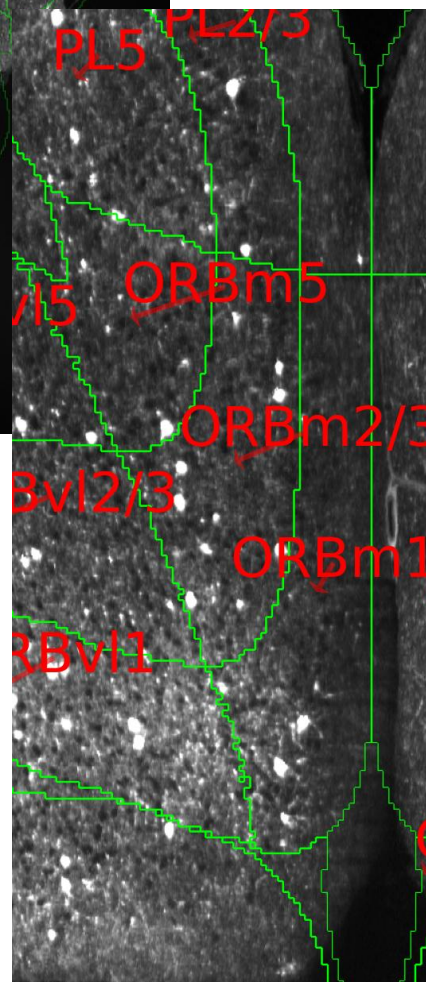
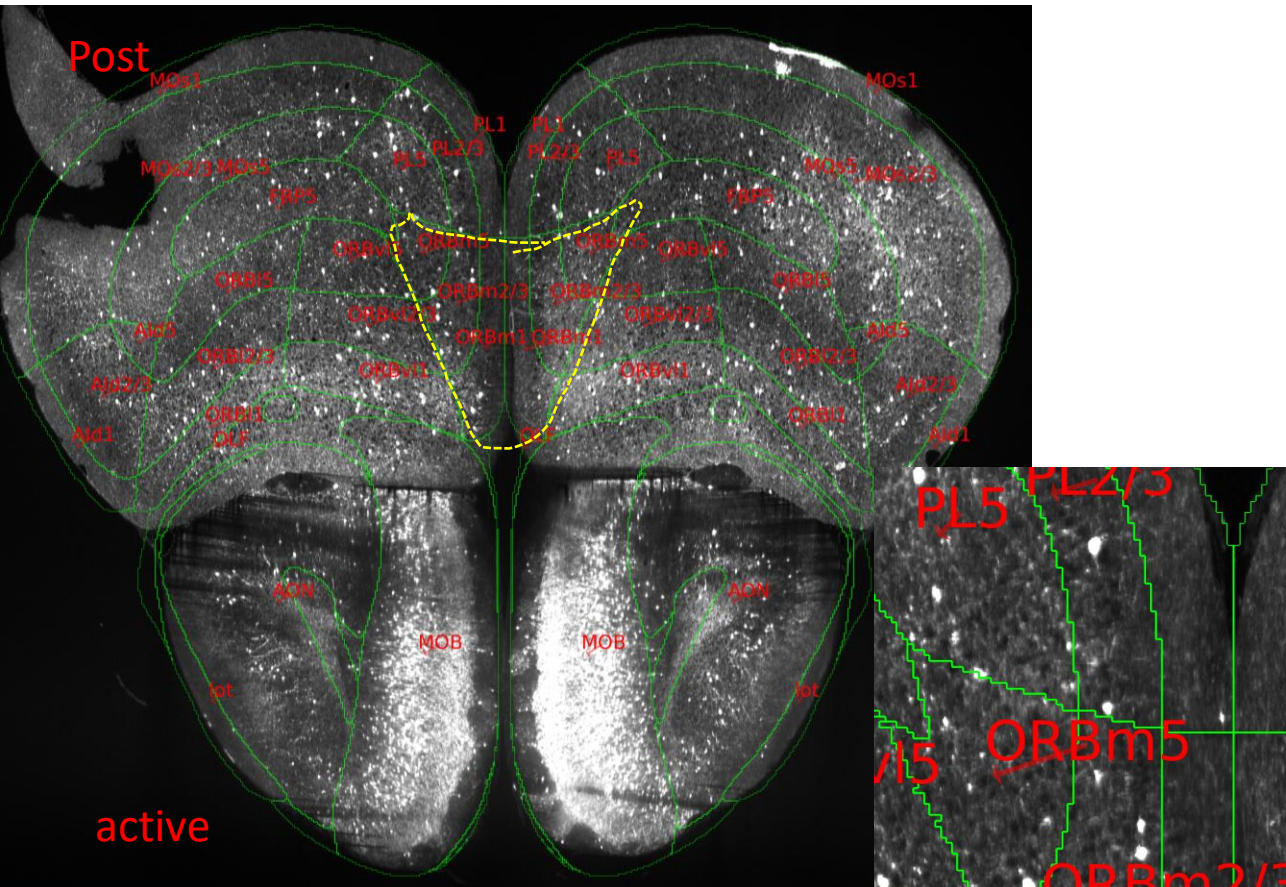


Passive

1000 μm

ORBm “Post” phase
Active mouse (left) vs Passive (right)

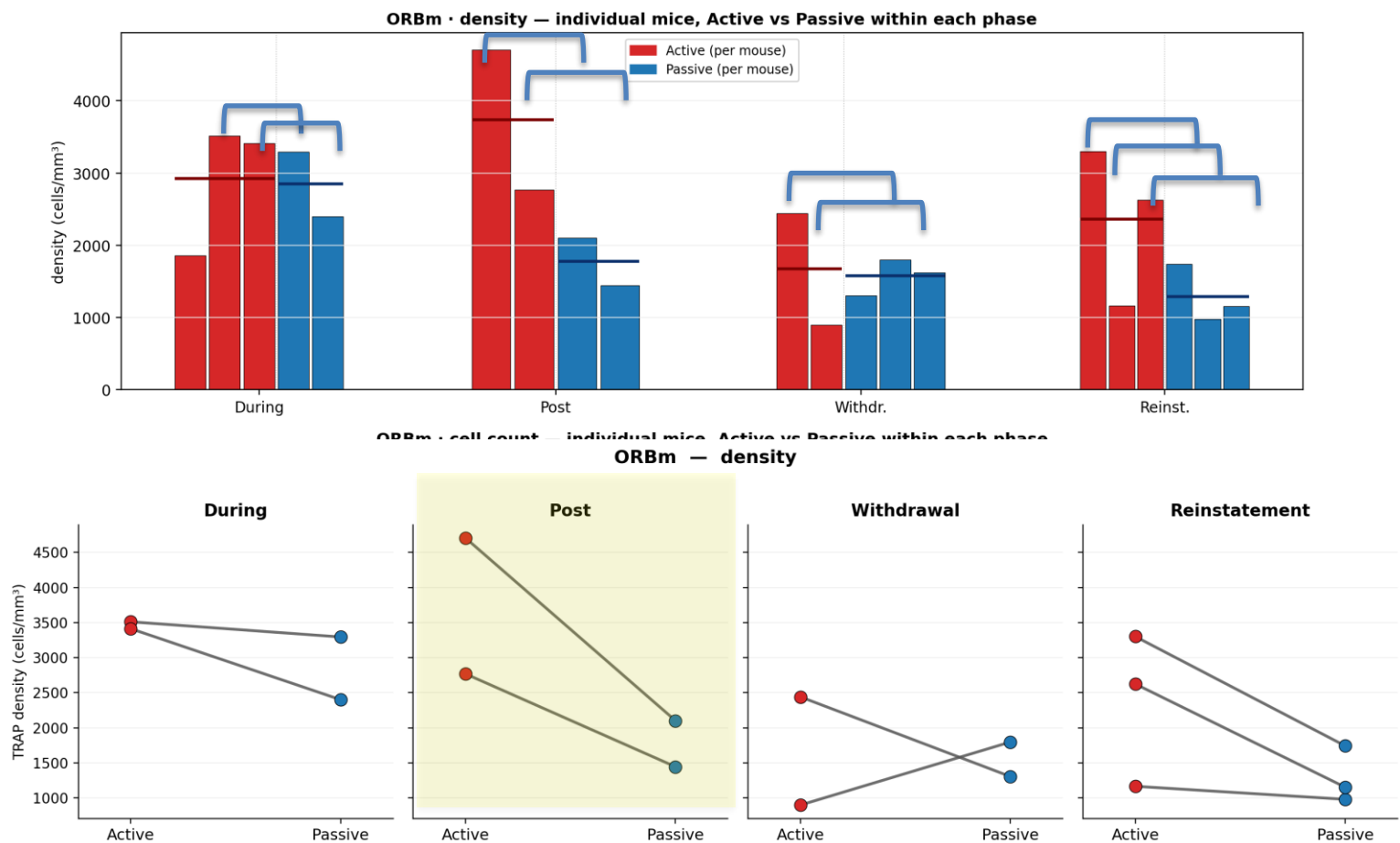




Finding A — ORBm within-phase (one bar = one mouse)

ORBm (Cluster 1) — within-phase Active vs Passive (one bar = one mouse)
horizontal line = group mean · comparison is WITHIN each phase (fair, same-phase mice)

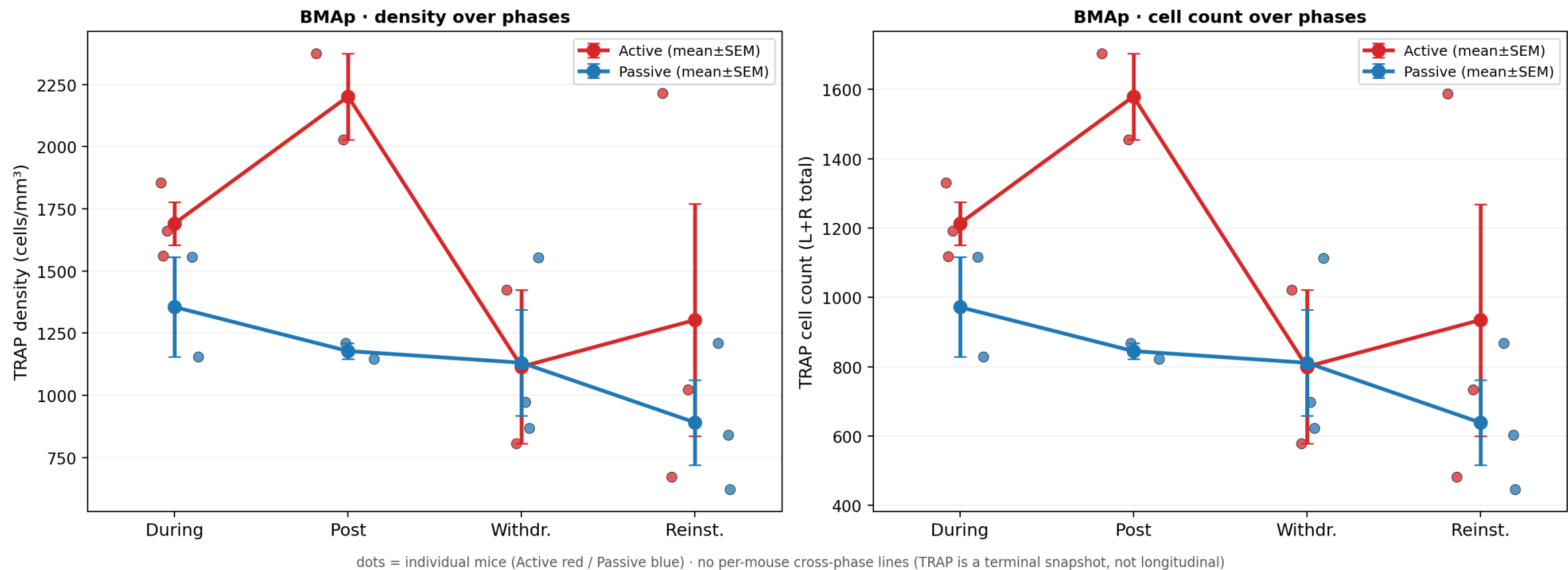
pair



Within each phase: every Active mouse vs every Passive mouse. Red = Active, blue = Passive.

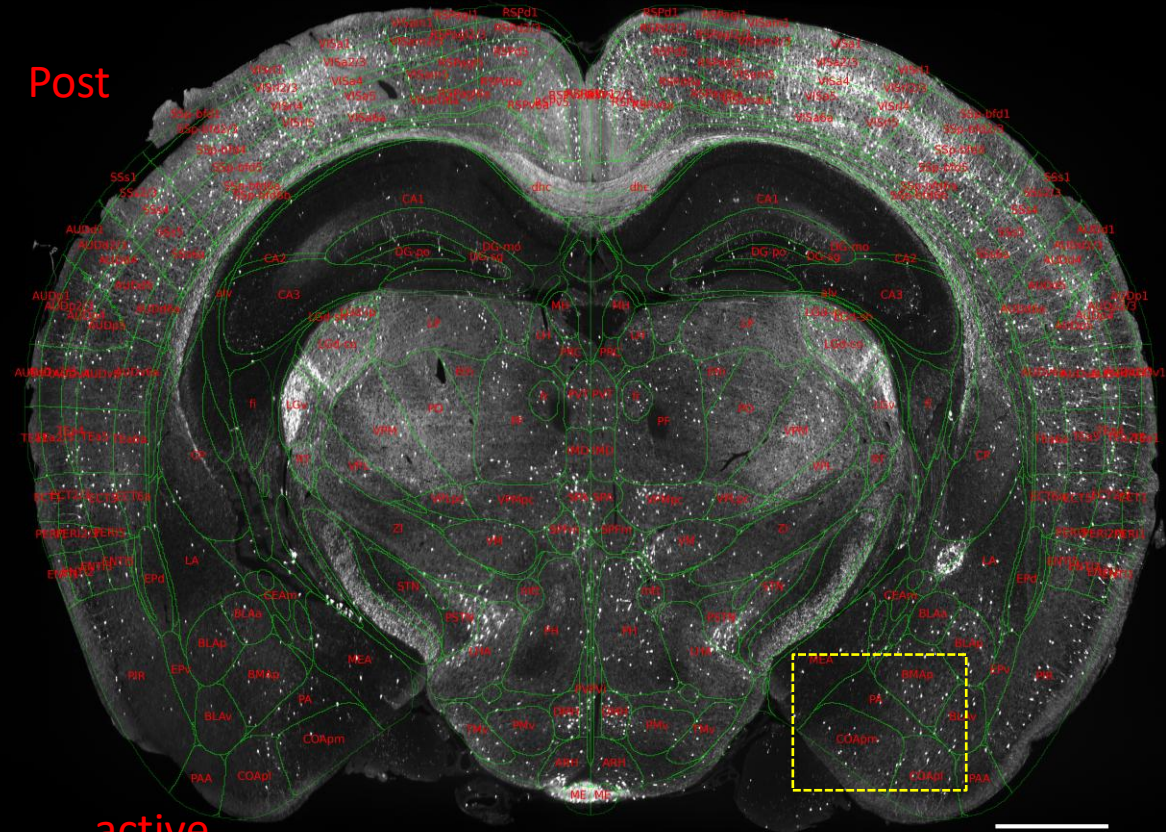
Finding A — BMAP zoom-in (density & cell count)

BMAP (Cluster 4) — individual mice (dots) + group mean±SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice



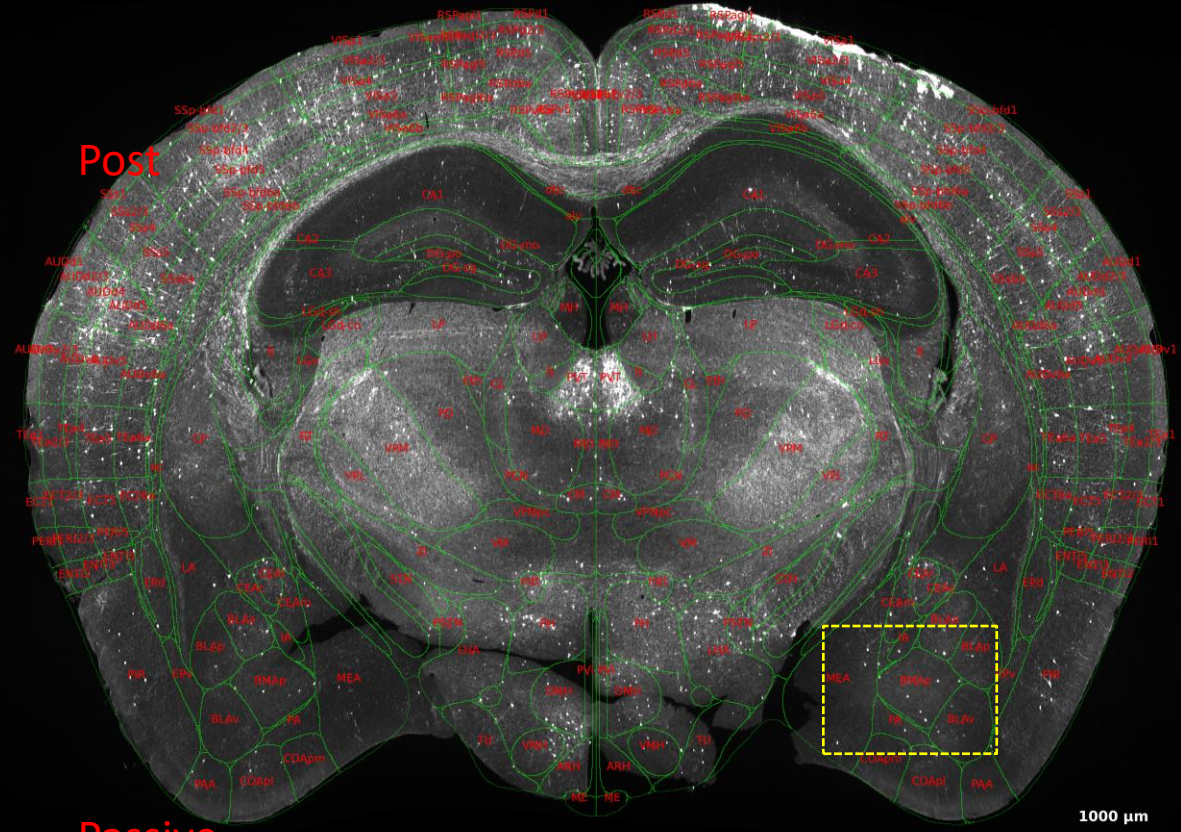
Group trajectories. Active peaks Post, dips Withdrawal, rebounds; Passive flat.

Post



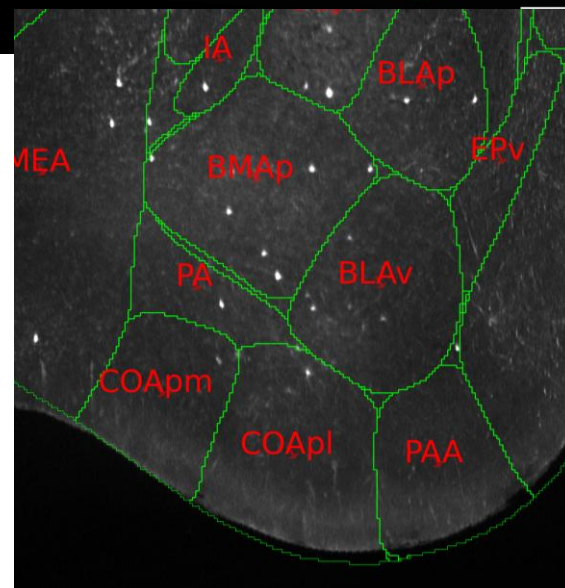
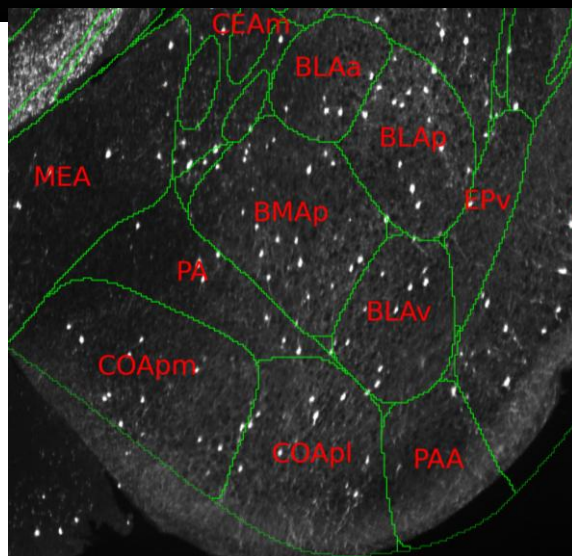
active

Post

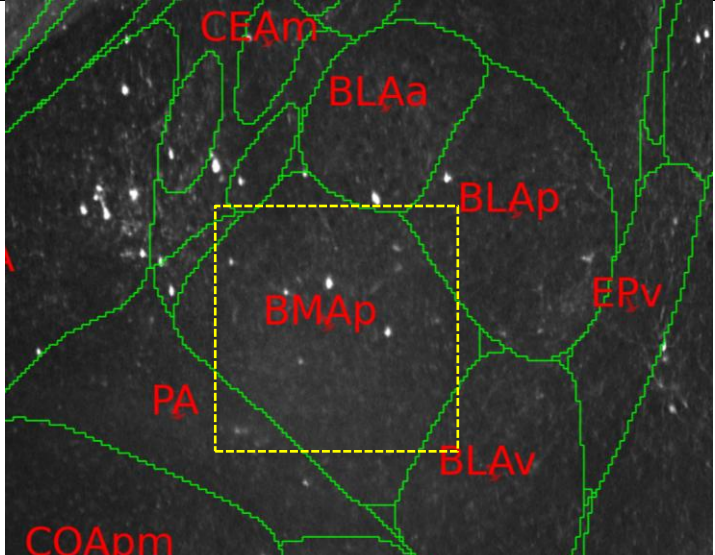
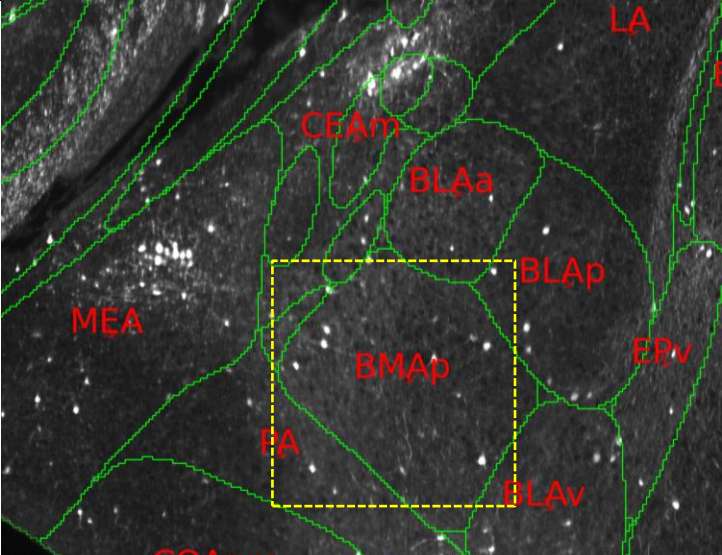
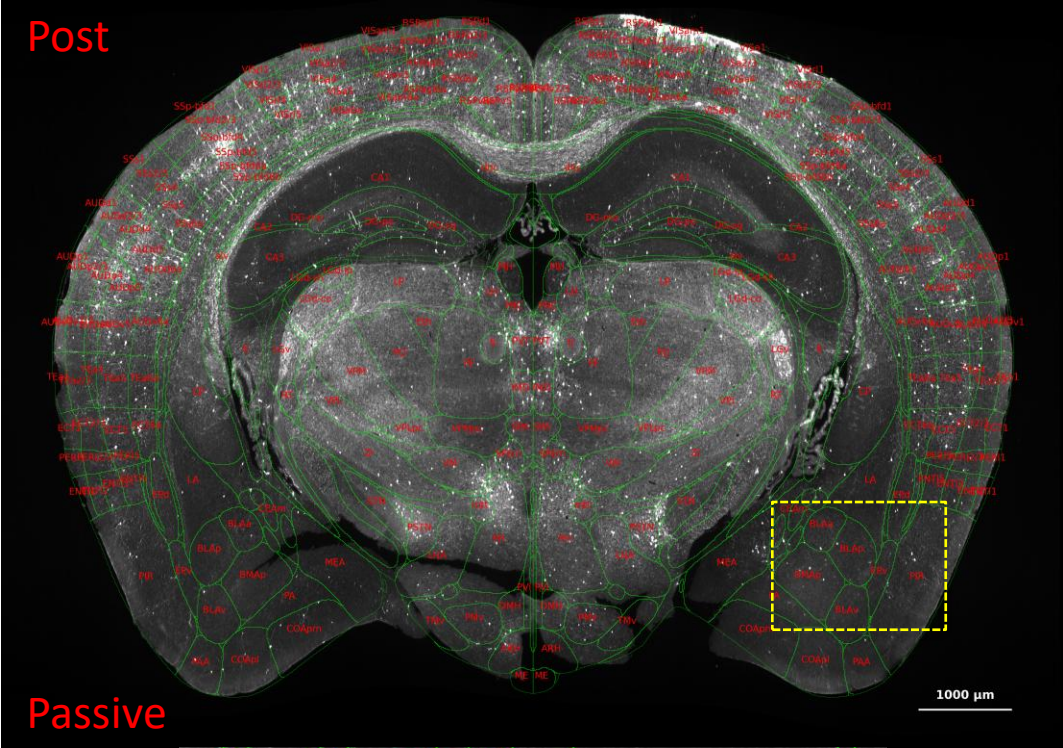
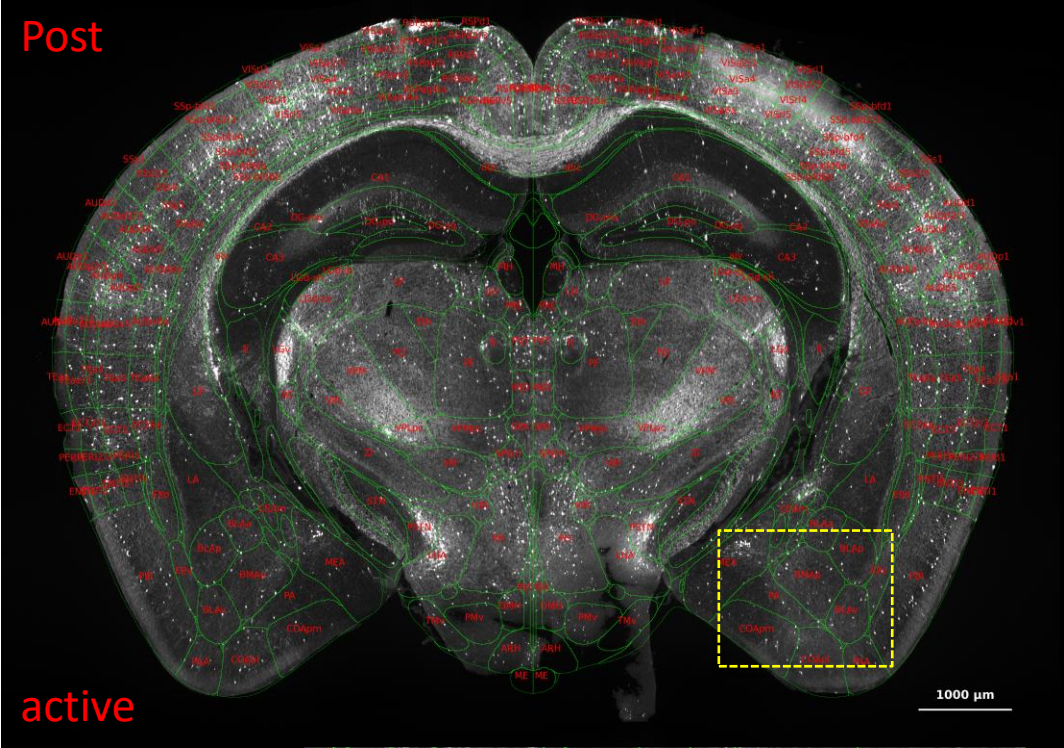


Passive

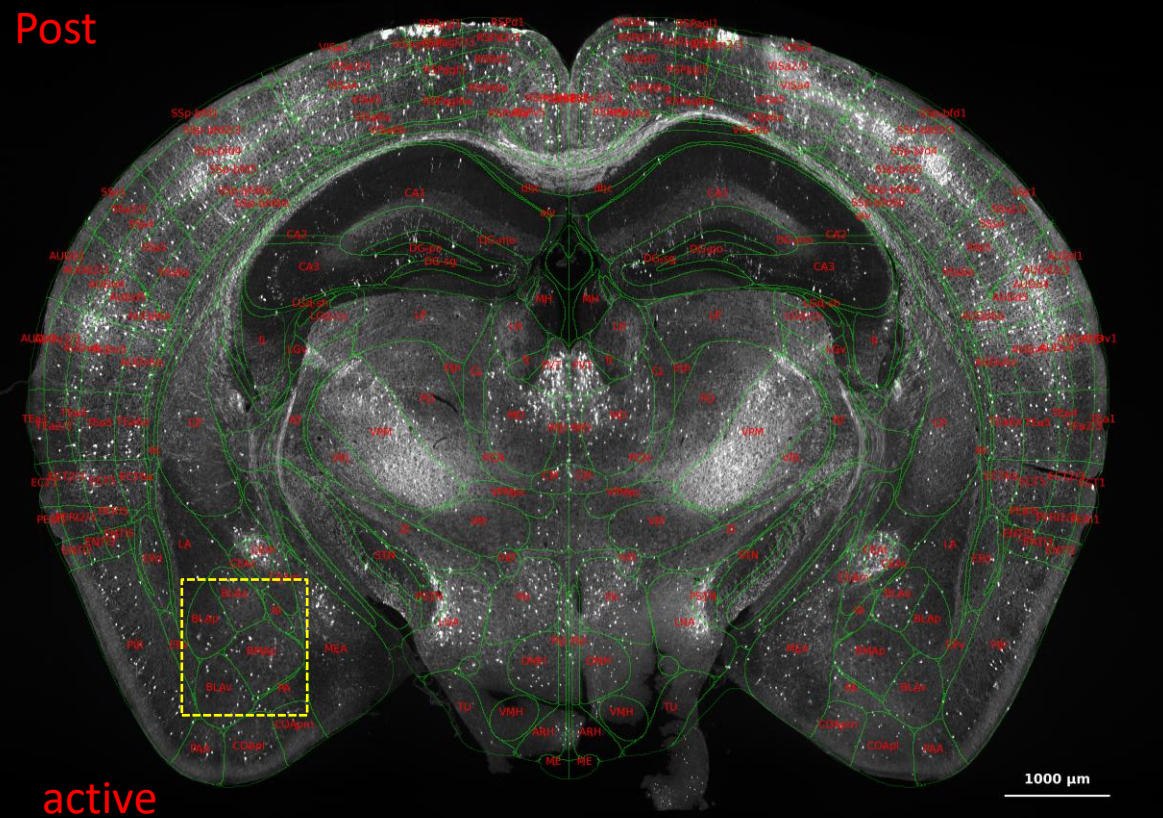
1000 μ m



BMap “Post” phase
Active mouse vs Passive mouse

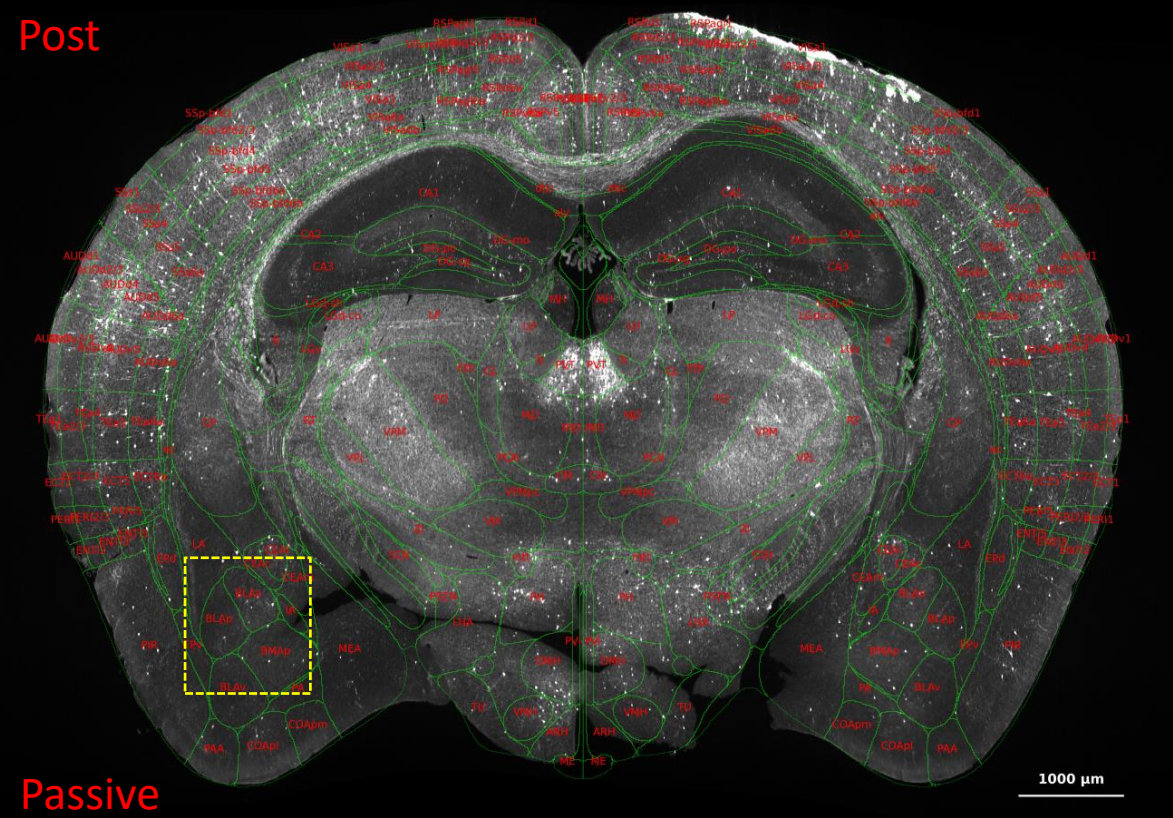


Post

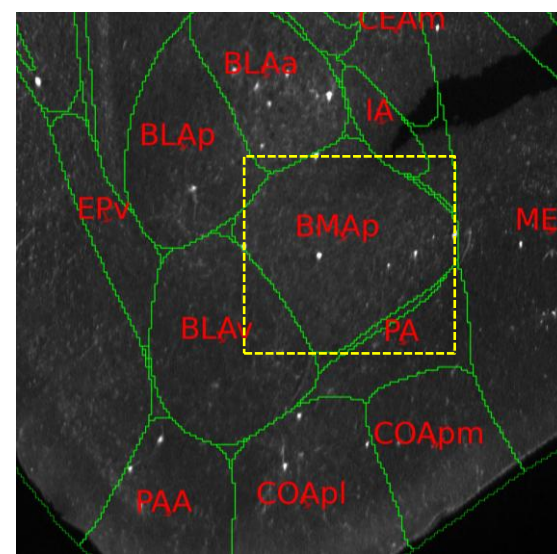
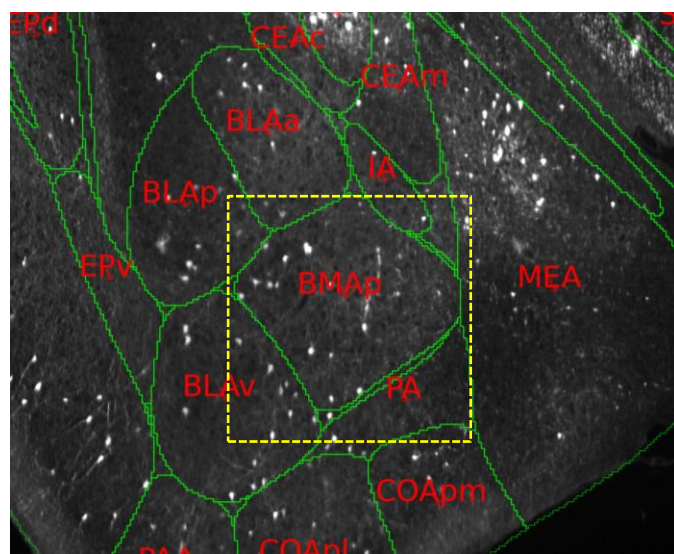


active

Post



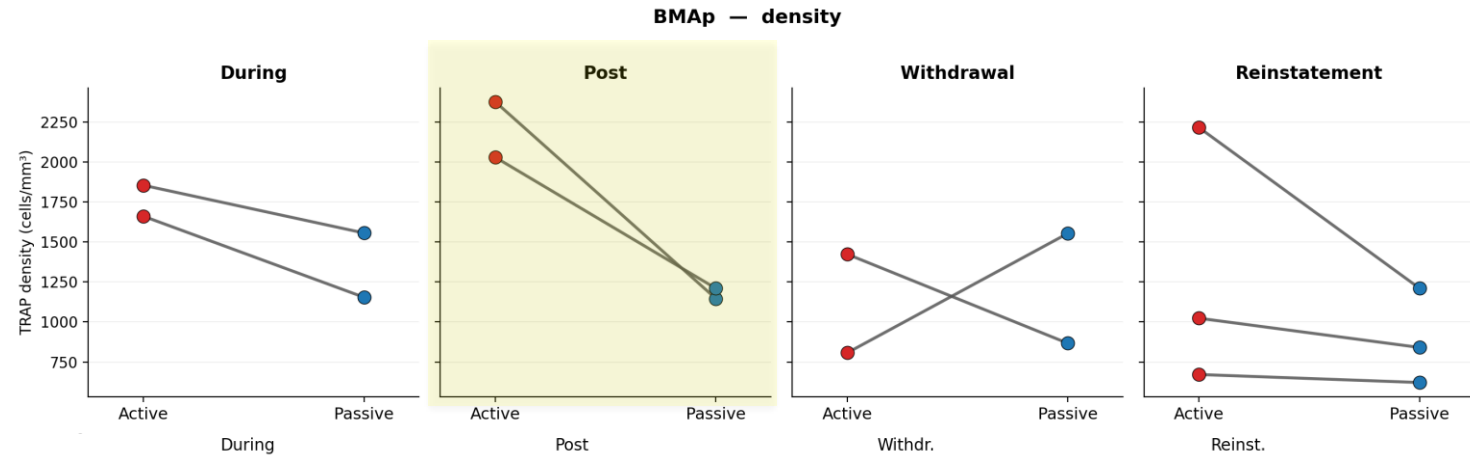
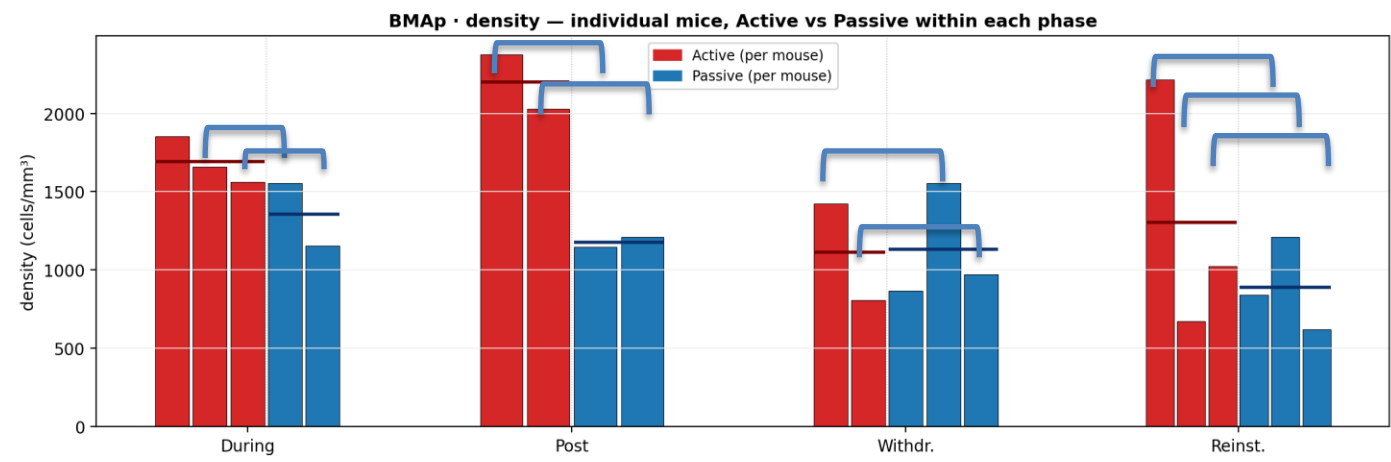
Passive



Finding A — BMAP within-phase (one bar = one mouse)

BMAP (Cluster 4) — within-phase Active vs Passive (one bar = one mouse)
horizontal line = group mean · comparison is WITHIN each phase (fair, same-phase mice)

pair



Within each phase: every Active mouse vs every Passive mouse. Red = Active, blue = Passive.

Whole-brain TRAP — another candidate stories

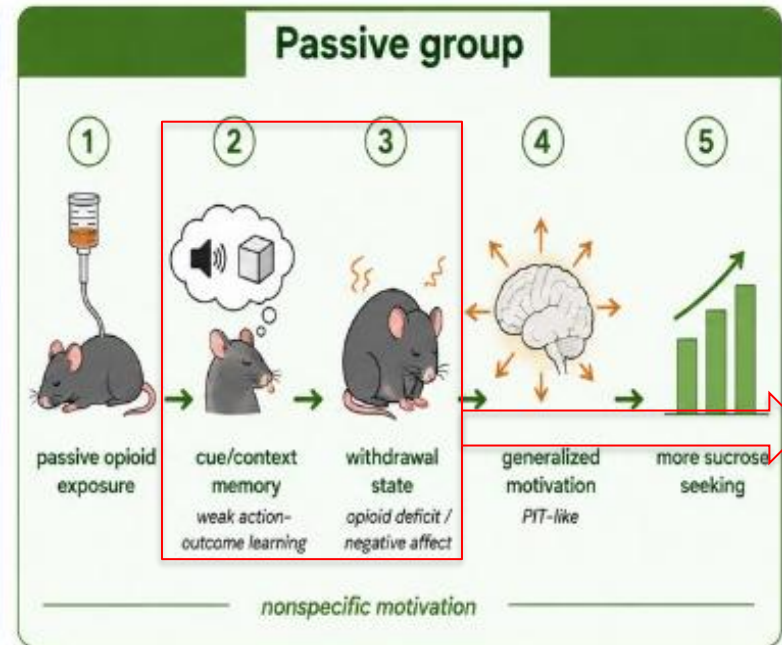
B) Passive withdrawal (patient-relevant)

second finding — why keep both

- FINDING B — COAa: Passive > Active at Post and Withdrawal (patient-relevant). Best anatomical fit in cluster 3 for cue/context + withdrawal-state → generalized motivation candidate (same numeric pattern also LGv, FC).
- COAa = best anatomical fit for Finding B schematic (cue/context + withdrawal-state → generalized motivation candidate).
- LGv (visual thalamus) & FC (hippocampal remnant) = same pattern, weaker anatomy for PIT / affect story.

Finding B — Why Passive matters (patient-relevant)

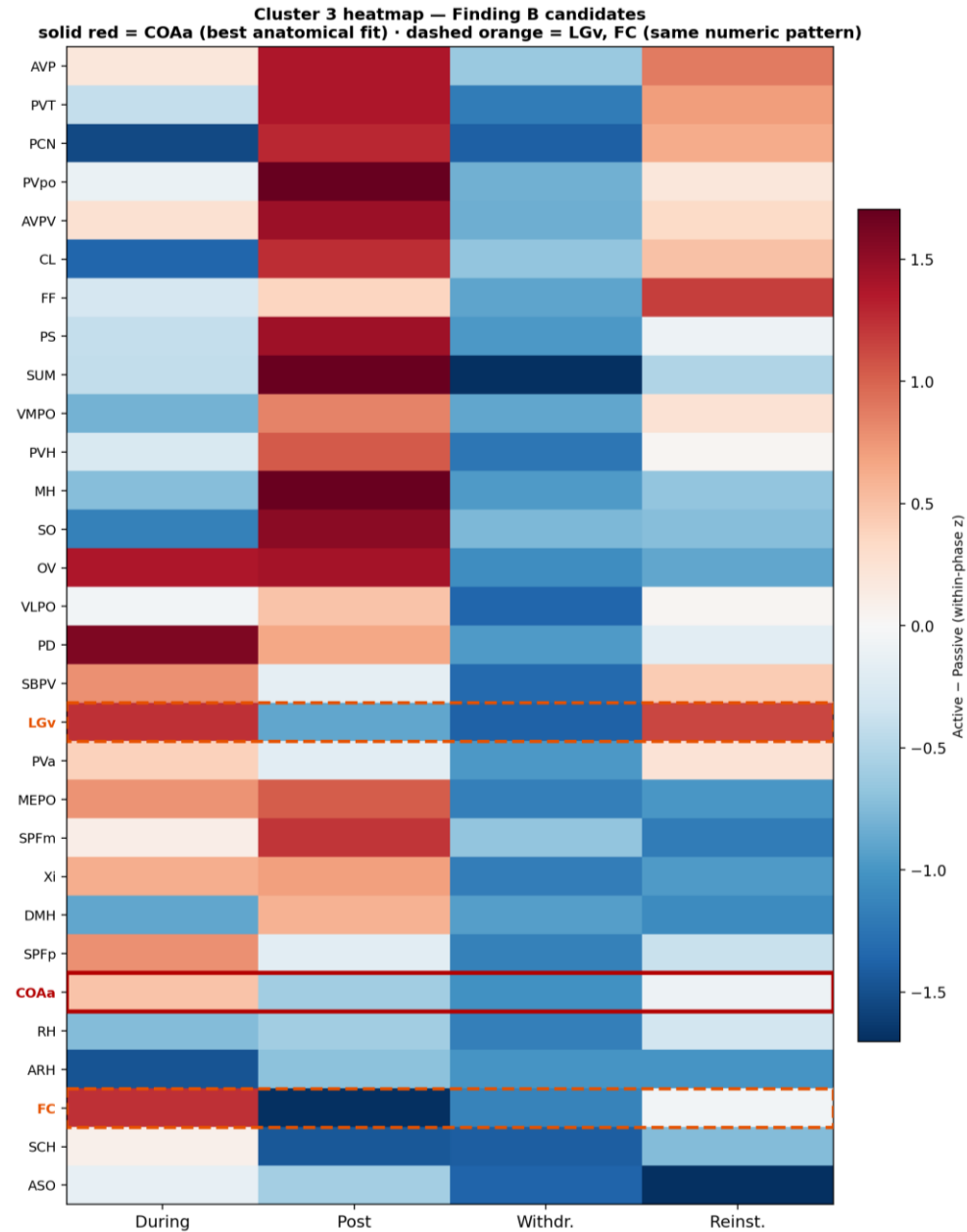
Why withdrawal affects sucrose seeking differently



Active = opioid-specific motivation; Passive = withdrawal-driven generalized motivation.

- Active: opioid-specific → less sucrose seeking in withdrawal.
 - Passive: generalized motivation → MORE sucrose seeking.
 - Behavior: withdrawal ↑ sucrose is Passive-only.
- Candidate: cue/context (Post) + withdrawal-state → generalized (PIT-like) motivation.
- Passive > Active at Post and Withdrawal in COAa (best anatomy in c3; also LGv/FC numerically).
- Patients often first exposed noncontingently → Passive is translational.

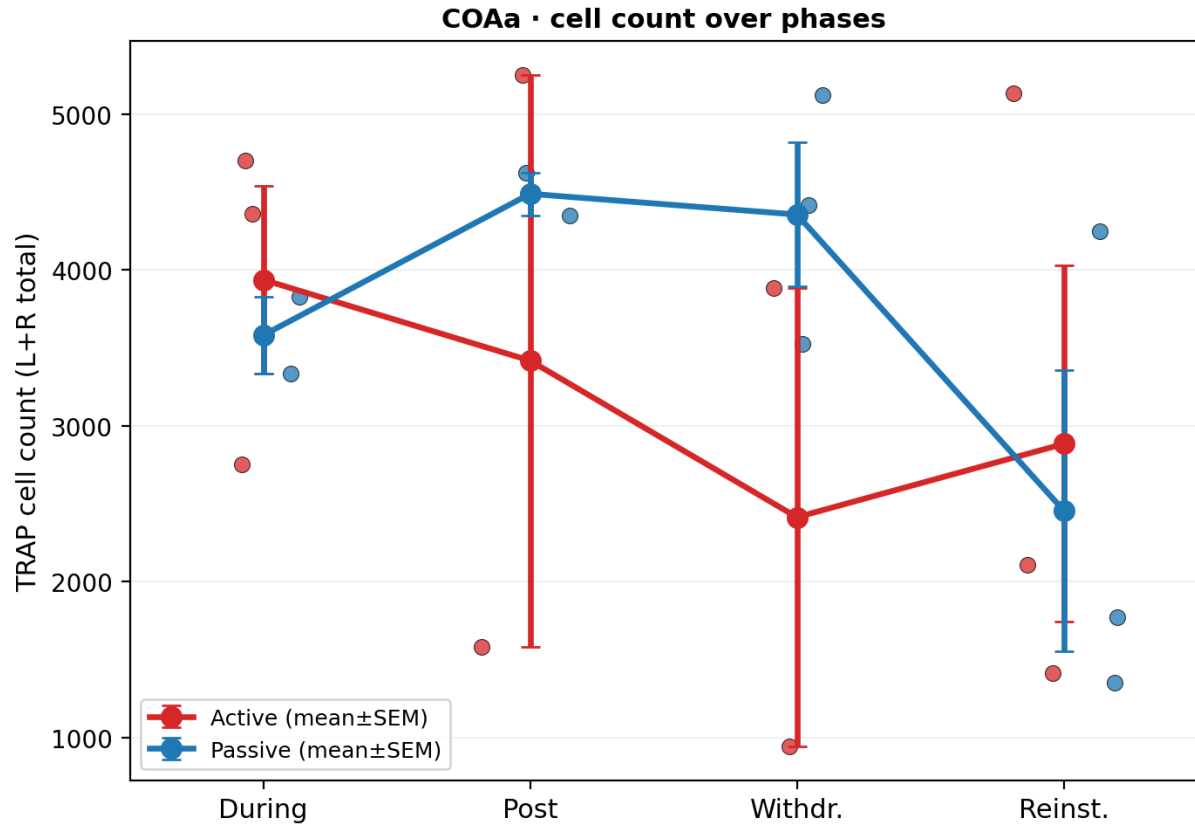
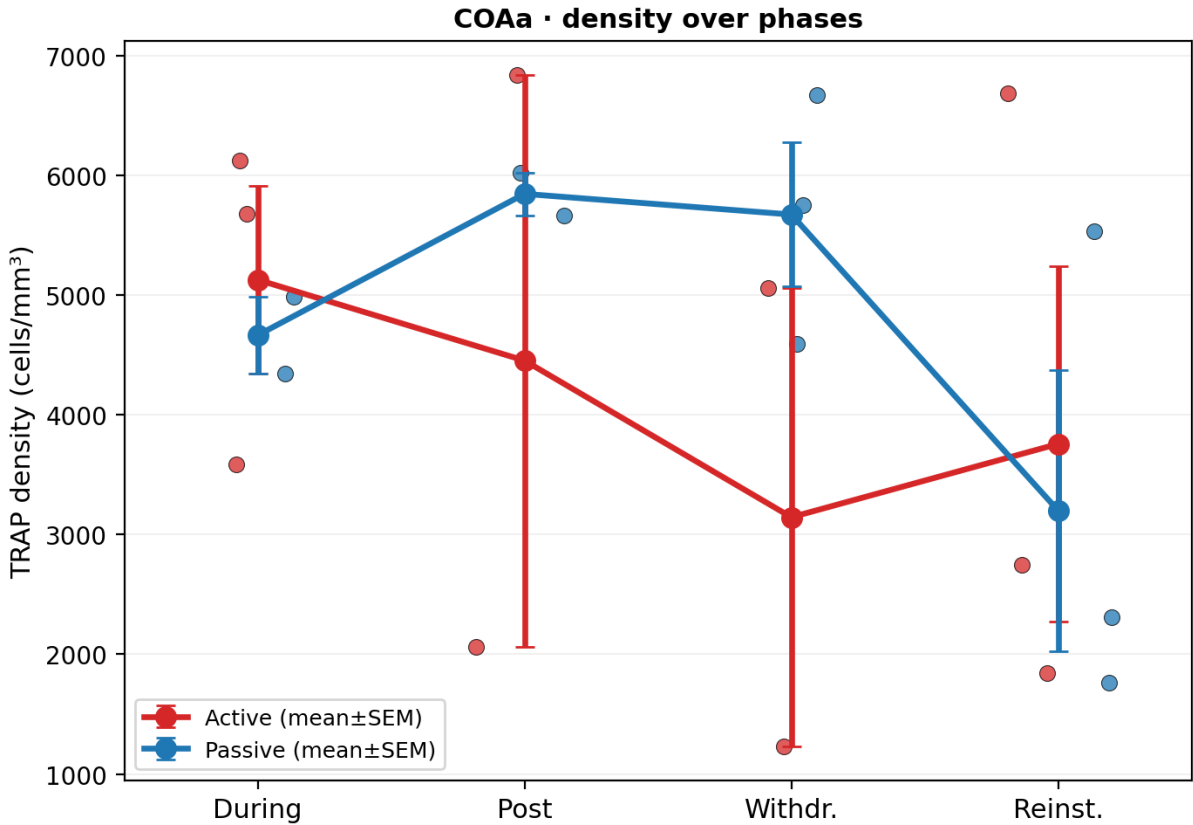
Cluster 3: heatmap — Finding B candidates



- Red box = COAa (chosen).
- Orange dashed = LGv, FC (numeric peers).
- Metric: Active – Passive (z).

Finding B — COAa zoom-in (density & cell count)

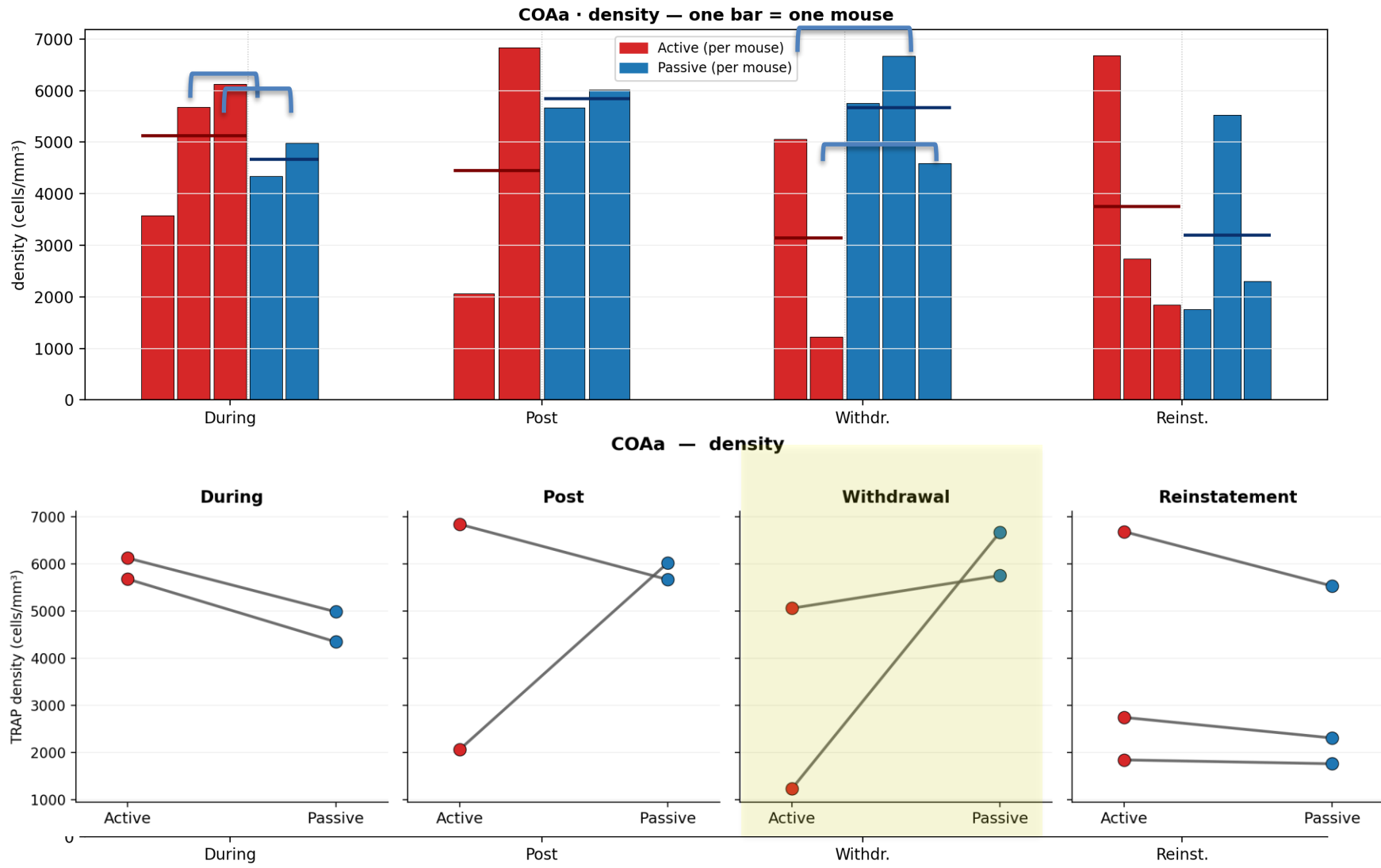
COAa (Cluster 3) — individual mice (dots) + group mean±SEM
terminal design: each mouse = one phase; lines connect group means across (different) mice



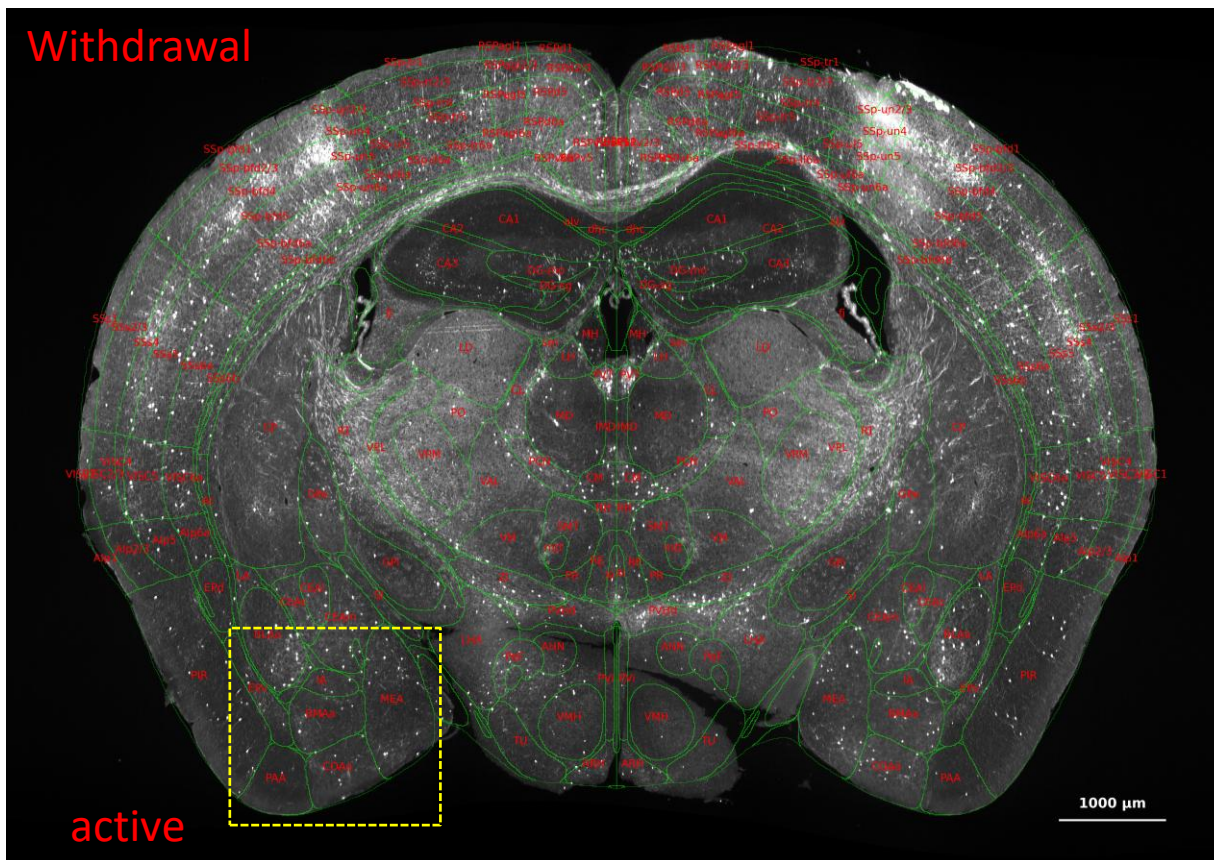
dots = individual mice (Active red / Passive blue) · no per-mouse cross-phase lines (TRAP is a terminal snapshot, not longitudinal)

Finding b — COAa within-phase (one bar = one mouse)

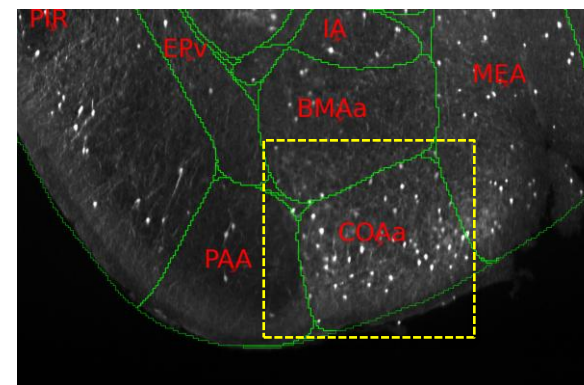
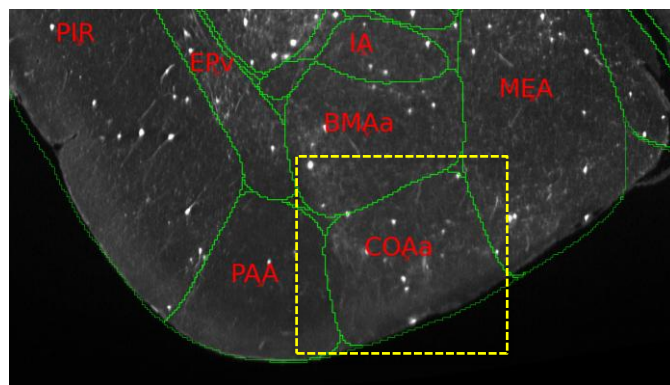
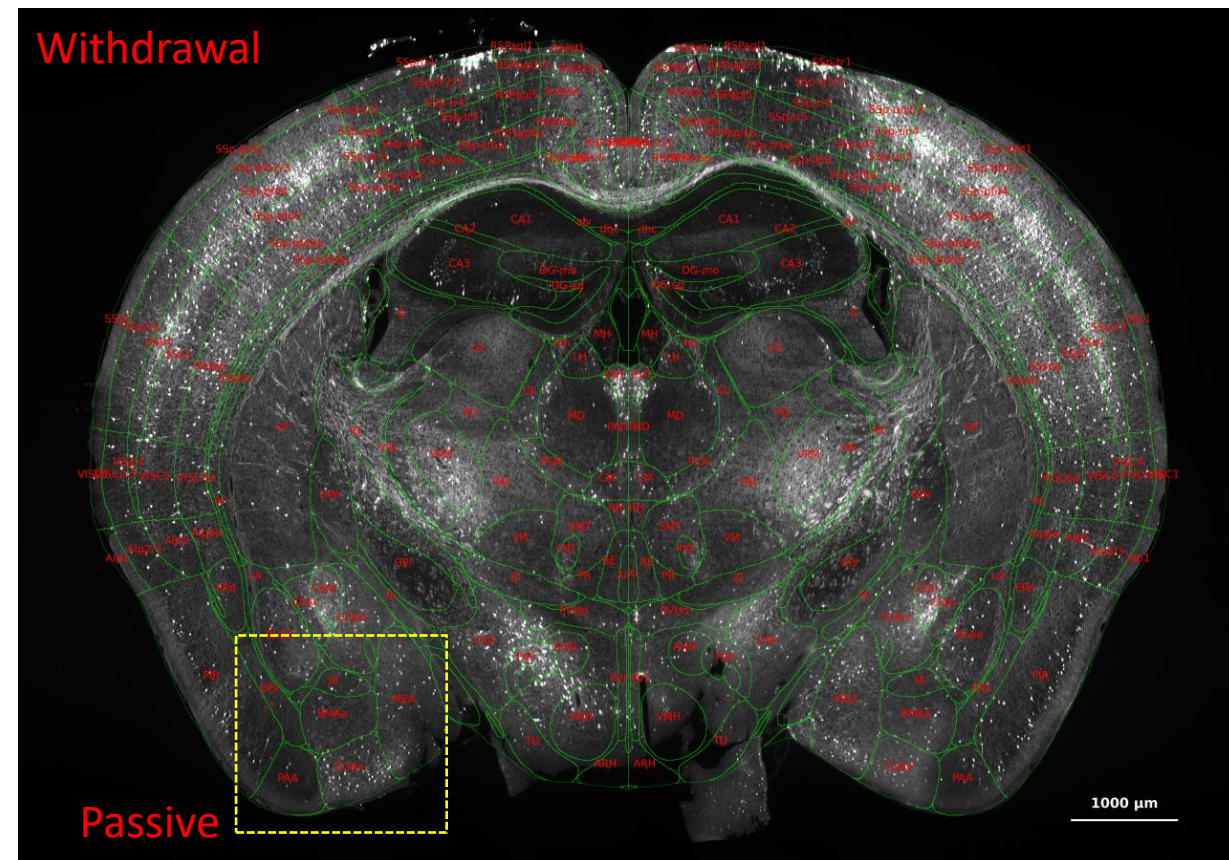
COAa (Cluster 3) — within-phase Active vs Passive
horizontal line = group mean · orange interest: Passive > Active at Post & Withdrawal



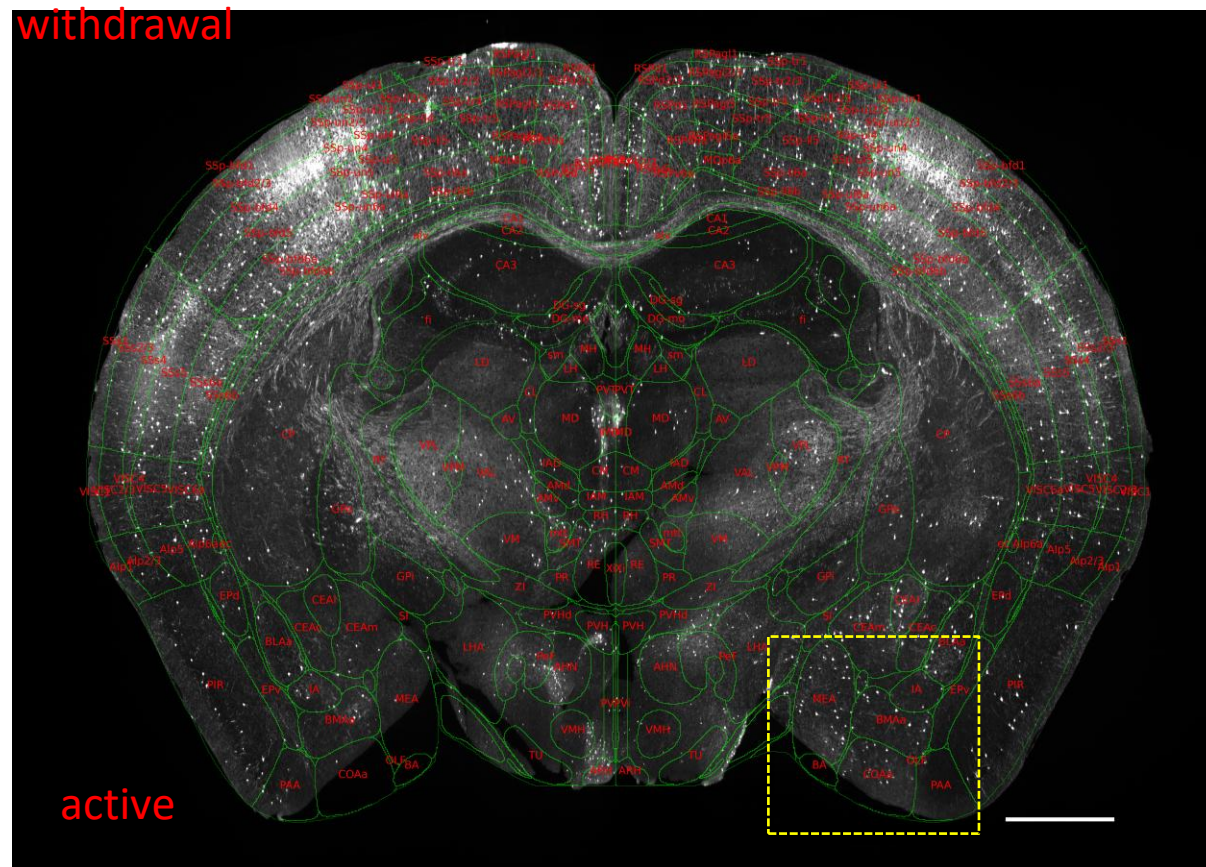
Withdrawal



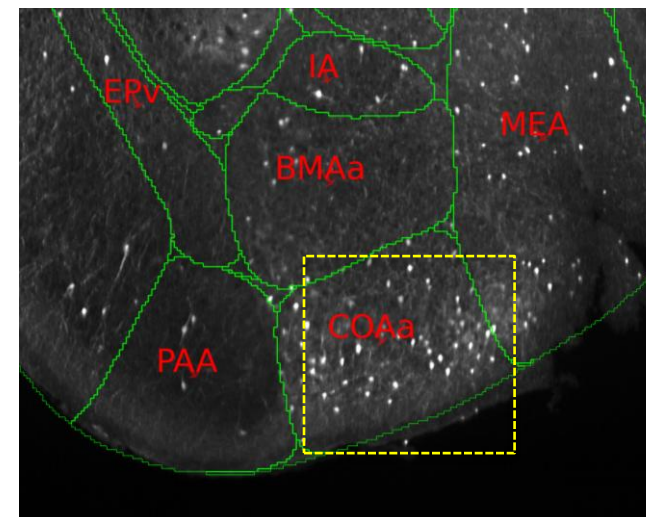
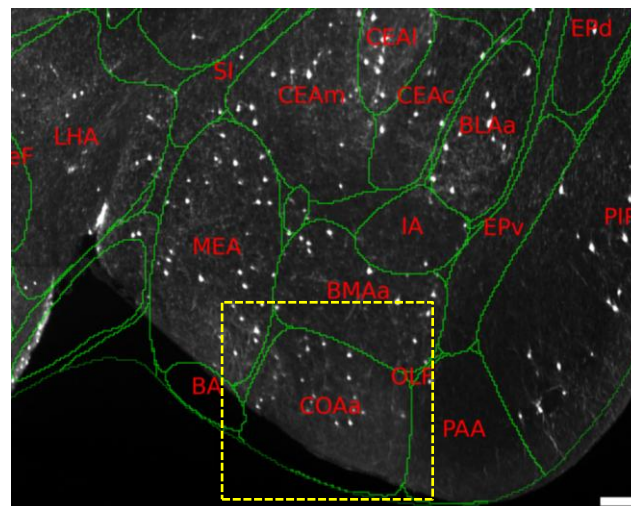
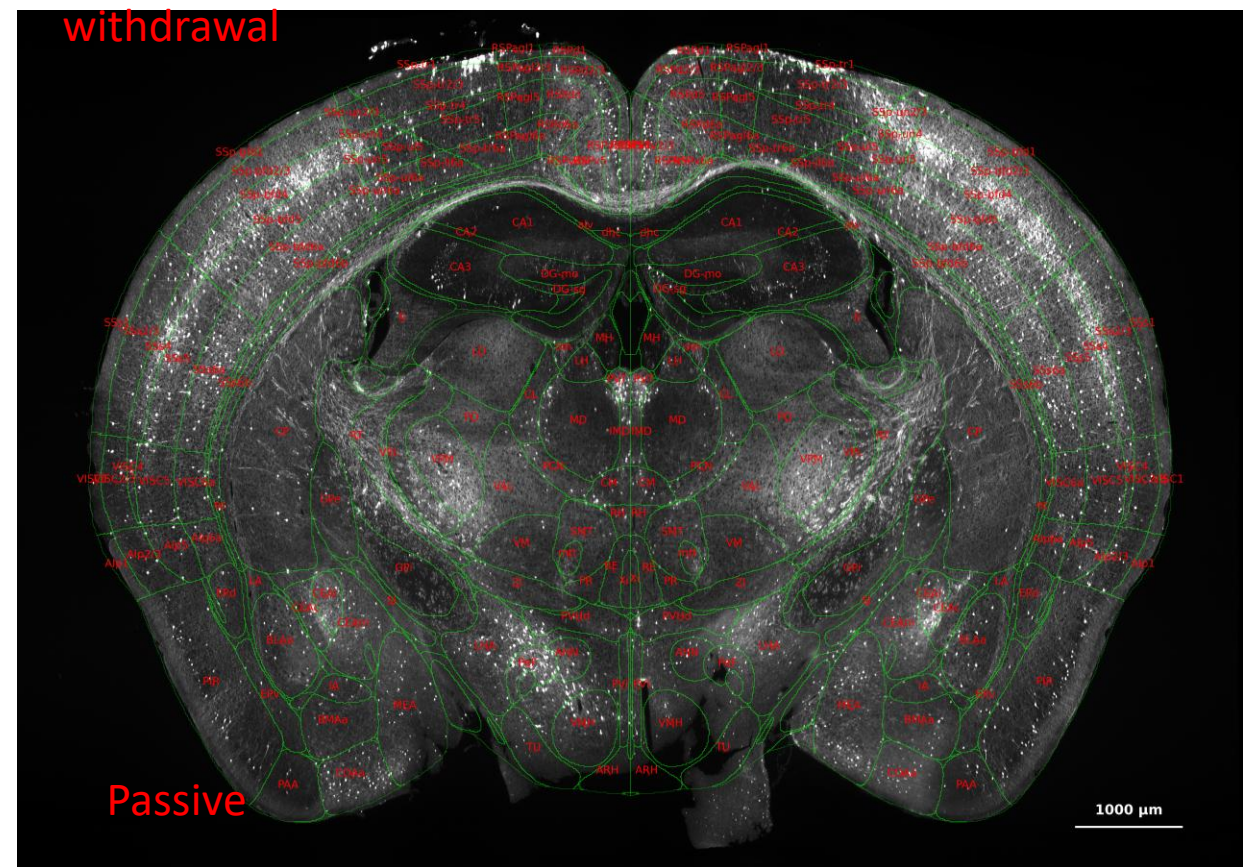
Withdrawal



withdrawal



withdrawal



Summary

- A) ORBm & BMAp — dual-role craving: opioid seeking + negative-affect in Withdrawal (Active >> Passive).
- B) COAa — Passive > Active at Post and Withdrawal (patient-relevant); best anatomical fit in c3
- Keep both → complementary Xenium / chemogenetic targets (A: Active craving ORBm/BMAp; B: Passive Post+WD bias COAa).

Activating COAa should bias toward broad reward seeking; activating ORBm/BMAp should bias toward opioid-related seeking. That's the dissociation I predict — still needs a causal test.